AFKEBOOKS
SINCE 2013

TABLE 288-4 Infectious Causes of Pulmonary Eosinophilia

Löffler Syndrome
Ascaris
Hookworm
Schistosomiasis
Heavy Parasite Burden
Strongyloidiasis
Direct Pulmonary Penetration
Paragonimiasis
Visceral larval migrans
Immunologic Response to Organisms in Lungs
Filariasis
Dirofilariasis
Cystic Disease
Echinococcus
Cysticercosis
Other Nonparasitic
Coccidioidomycosis
Basidiobolomycosis
Paracoccidioidomycosis
Tuberculosis

Source: Adapted from P Akuthota, PF Weller: Clin Microbiol Rev 25:649, 2012.

GLOBAL CONSIDERATIONS

In the United States, drug-induced eosinophilic pneumonias are the most common cause of eosinophilic pulmonary infiltrates. A travel history or evidence of recent immigration should prompt the consideration of parasite-associated disorders. Tropical eosinophilia is usually caused by filarial infection; however, eosinophilic pneumonias also occur with other parasites such as *Ascaris* spp., *Ancylostoma* spp., *Toxocara* spp., and *Strongyloides stercoralis*. Tropical eosinophilia due to *Wuchereria bancrofti* or *Wuchereria malayi* occurs most commonly in southern Asia, Africa, and South America and is treated successfully with diethylcarbamazine. In the United States, *Strongyloides* is endemic to the southeastern and Appalachian regions.

FURTHER READING

AKUTHOTA P, WELLER PF: Eosinophilic pneumonias. Clin Microbiol Rev 25:649, 2012.

COTTIN V: Eosinophilic lung diseases. Clin Chest Med 37:535, 2016.

VASAKOVA M et al: Hypersensitivity pneumonitis: Perspectives in diagnosis and management. Am J Respir Crit Care Med 196:680, 2017.

WECHSLER ME et al: Mepolizumab or placebo for eosinophilic granulomatosis with polyangiitis. N Engl J Med 376:1921, 2017.

289 Occupational and Environmental Lung Disease

John R. Balmes

Occupational and environmental lung diseases are difficult to distinguish from those of nonenvironmental origin. Virtually all major categories of pulmonary disease can be caused by environmental agents, and environmentally related disease usually presents clinically in a manner indistinguishable from that of disease not caused by such agents. In addition, the etiology of many diseases may be multifactorial; occupational and environmental factors may interact with other factors (such as smoking and genetic risk). It is often only after a careful exposure history is taken that the underlying workplace or general environmental exposure is uncovered.

Why is knowledge of occupational or environmental etiology so important? Patient management and prognosis are affected significantly by such knowledge. For example, patients with occupational asthma or hypersensitivity pneumonitis often cannot be managed adequately without cessation of exposure to the offending agent. Establishment of cause may have significant legal and financial implications for a patient who no longer can work in his or her usual job. Other exposed people may be identified as having the disease or prevented from getting it. In addition, new associations between exposure and disease may be identified (e.g., nylon flock worker's lung disease and diacetyl-induced bronchiolitis obliterans).

Although the exact proportion of lung disease due to occupational and environmental factors is unknown, a large number of individuals are at risk. For example, 15–20% of the burden of adult asthma and chronic obstructive pulmonary disease (COPD) has been estimated to be due to occupational factors.

HISTORY AND EXPOSURE ASSESSMENT

The patient's history is of paramount importance in assessing any potential occupational or environmental exposure. Inquiry into specific work practices should include questions about the specific contaminants involved, the presence of visible dusts, chemical odors, the size and ventilation of workspaces, the use of respiratory protective equipment, and whether coworkers have similar complaints. The temporal association of exposure at work and symptoms may provide clues to occupation-related disease. In addition, the patient must be questioned about alternative sources of exposure to potentially toxic agents, including hobbies, home characteristics, exposure to secondhand smoke, and proximity to traffic or industrial facilities. Short-term and long-term exposures to potential toxic agents in the distant past also must be considered.

Workers in the United States have the right to know about potential hazards in their workplaces under federal Occupational Safety and Health Administration (OSHA) regulations. Employers must provide specific information about potential hazardous agents in products being used through Safety Data Sheets as well as training in personal protective equipment and environmental control procedures. However, the introduction of new processes and/or new chemical compounds may change exposure significantly, and often only the employee on the production line is aware of the change. For the physician caring for a patient with a suspected work-related illness, a visit to the work site can be very instructive. Alternatively, an affected worker can request an inspection by OSHA. If reliable environmental sampling data are available, that information should be used in assessing a patient's exposure. Because chronic diseases may result from exposure over many years, current environmental measurements should be combined with work histories to arrive at estimates of past exposure.

LABORATORY TESTS

Exposures to inorganic and organic dusts can cause interstitial lung disease that presents with a restrictive pattern and a decreased diffusing capacity (Chap. 285). Similarly, exposures to a number of dusts or chemical agents may result in occupational asthma or COPD that is characterized by airway obstruction. Measurement of change in forced expiratory volume in 1 s (FEV_1) before and after a working shift can be used to detect an acute bronchoconstrictive response.

The chest radiograph is useful in detecting and monitoring the pulmonary response to mineral dusts, certain metals, and organic dusts capable of inducing hypersensitivity pneumonitis. The International Labour Organisation (ILO) International Classification of Radiographs of Pneumoconioses classifies chest radiographs by the nature and size of opacities seen and the extent of involvement of the parenchyma. In general, small rounded opacities are seen in silicosis or coal worker's pneumoconiosis, and small linear opacities are seen in asbestosis. Although useful for epidemiologic studies and screening large numbers of workers, the ILO system can be problematic when

AFKEBOOKS
SINCE 2013

applied to an individual worker's chest radiograph. With dusts causing rounded opacities, the degree of involvement on the chest radiograph may be extensive, whereas pulmonary function may be only minimally impaired. In contrast, in pneumoconiosis causing linear, irregular opacities like those seen in asbestosis, the radiograph may lead to underestimation of the severity of the impairment until relatively late in the disease. For patients with a history of asbestos exposure, conventional computed tomography (CT) is more sensitive for the detection of pleural thickening, and high-resolution CT (HRCT) improves the detection of asbestosis.

Other procedures that may be of use in identifying the role of environmental exposures in causing lung disease include skin prick testing or specific IgE antibody titers for evidence of immediate hypersensitivity to agents capable of inducing occupational asthma (e.g., flour antigens in bakers), specific IgG precipitating antibody titers for agents capable of causing hypersensitivity pneumonitis (e.g., pigeon antigen in bird handlers), and assays for specific cell-mediated immune responses (e.g., beryllium lymphocyte proliferation testing in nuclear workers or tuberculin skin testing in health care workers). Sometimes a bronchoscopy to obtain transbronchial biopsies of lung tissue may be required for histologic diagnosis (chronic beryllium disease [CBD]). Rarely, video-assisted thoracoscopic surgery to obtain a larger sample of lung tissue may be required to determine the specific diagnosis of environmentally induced lung disease (hypersensitivity pneumonitis or giant cell interstitial pneumonitis due to cobalt exposure).

DETERMINANTS OF INHALATIONAL EXPOSURE

The chemical and physical characteristics of inhaled agents affect both the dose and the site of deposition in the respiratory tract. Water-soluble gases such as ammonia and sulfur dioxide are absorbed in the lining fluid of the upper and proximal airways and thus tend to produce irritative and bronchoconstrictive responses. In contrast, nitrogen dioxide and phosgene, which are less soluble, may penetrate to the bronchioles and alveoli in sufficient quantities to produce acute chemical pneumonitis.

Particle size of air contaminants must also be considered. Because of their settling velocities in air, particles >10–15 μm in diameter do not penetrate beyond the nose and throat. Particles <10 μm in size are deposited below the larynx. These particles are divided into three size fractions on the basis of their size characteristics and sources. Particles ~2.5–10 μm (coarse-mode fraction) contain crustal elements such as silica, aluminum, and iron. These particles mostly deposit relatively high in the tracheobronchial tree. Although the total mass of an ambient sample is dominated by these larger respirable particles, the number of particles, and therefore the surface area on which potential toxic agents can deposit and be carried to the lower airways, is dominated by particles <2.5 μm (fine-mode fraction). These fine particles are created primarily by the burning of fossil fuels or high-temperature industrial processes resulting in condensation products from gases, fumes, or vapors. The smallest particles, those <0.1 μm in size, represent the ultrafine fraction and make up the largest number of particles; they tend to remain in the airstream and deposit in the lung only on a random basis as they come into contact with the alveolar walls. If they do deposit, however, particles of this size range may penetrate into the circulation and be carried to extrapulmonary sites. New technologies create particles of this size ("nanoparticles") for use in many commercial applications. Besides the size characteristics of particles and the solubility of gases, the actual chemical composition, mechanical properties, and immunogenicity or infectivity of inhaled material determine in large part the nature of the diseases found among exposed persons.

OCCUPATIONAL EXPOSURES AND PULMONARY DISEASE

Table 289-1 provides broad categories of exposure in the workplace and diseases associated with chronic exposure in those industries.

ASBESTOS-RELATED DISEASES

Asbestos is a generic term for several different mineral silicates, including chrysolite, amosite, anthophyllite, and crocidolite. In addition to

TABLE 289-1 Categories of Occupational Exposure and Associated Respiratory Conditions

OCCUPATIONAL EXPOSURES	NATURE OF RESPIRATORY RESPONSES	COMMENT
Inorganic Dusts		
Asbestos: mining, processing, construction, ship repair	Fibrosis (asbestosis), pleural disease, cancer, mesothelioma	Virtually all new mining and construction with asbestos done in developing countries
Silica: mining, stone cutting, sandblasting, quarrying, artificial stone manufacture and installation	Fibrosis (silicosis), progressive massive fibrosis (PMF), cancer, tuberculosis, chronic obstructive pulmonary disease (COPD)	Improved protection in United States; persistent risk in developing countries
Coal dust: mining	Fibrosis (coal worker's pneumoconiosis), PMF, COPD	Risk persists in certain areas of United States, increasing in countries where new mines open
Beryllium: processing alloys for nuclear power and weapons, aerospace, and electronics	Acute pneumonitis (rare), chronic granulomatous disease, lung cancer (highly suspect)	Risk in high-tech industries persists
Other metals: aluminum, chromium, cobalt, nickel, titanium, tungsten carbide, or "hard metal" (contains cobalt)	Wide variety of conditions from acute pneumonitis to lung cancer and asthma	New diseases appear with new process development
Organic Dusts		
Cotton dust: milling, processing	Byssinosis (an asthma-like syndrome), chronic bronchitis, COPD	Increasing risk in developing countries with drop in United States as jobs shift overseas
Grain dust: elevator agents, dock workers, milling, bakers	Asthma, chronic bronchitis, COPD	Risk shifting more to migrant labor pool
Other agricultural dusts: fungal spores, vegetable products, insect fragments, animal dander, bird and rodent feces, endotoxins, microorganisms, pollens	Hypersensitivity pneumonitis (farmer's lung), asthma, chronic bronchitis	Important in migrant labor pool but also resulting from in-home exposures
Toxic chemicals: wide variety of industries; see Table 289-2	Asthma, chronic bronchitis, COPD, hypersensitivity pneumonitis, pneumoconiosis, and cancer	Reduced risk with recognized hazards; increasing risk for developing countries where controlled labor practices are less stringent
Other Environmental Agents		
Uranium and radon daughters, secondhand tobacco smoke, polycyclic aromatic hydrocarbons (PAHs), biomass smoke, diesel exhaust, welding fumes, wood finishing	Occupational exposures estimated to contribute to up to 10% of all lung cancers; chronic bronchitis, COPD, and fibrosis	In-home exposures important; in developing countries, biomass smoke is a major risk factor for COPD among women in these countries

AFKEBOOKS
SINCE 2013

workers involved in the production of asbestos products (mining, milling, and manufacturing), many workers in the shipbuilding and construction trades, including pipe fitters and boilermakers, were occupationally exposed because asbestos was widely used during the twentieth century for its thermal and electrical insulation properties. Asbestos also was used in the manufacture of fire-resistant textiles, in cement and floor tiles, and in friction materials such as brake and clutch linings.

Exposure to asbestos is not limited to persons who directly handle the material. Cases of asbestos-related diseases have been encountered in individuals with only bystander exposure, such as painters and electricians who worked alongside insulation workers in a shipyard. Community exposure resulted from the use of asbestos-containing mine and mill tailings as landfill, road surface, and playground material (e.g., Libby, MT, the site of a vermiculite mine in which the ore was contaminated with asbestos). Finally, exposure can occur from the disturbance of naturally occurring asbestos (e.g., from increasing residential development in the foothills of the Sierra Mountains in California).

Asbestos has largely been replaced in the developed world with synthetic mineral fibers such as fiberglass and refractory ceramic fibers, but it continues to be used in the developing world. The major health effects from exposure to asbestos are pleural and pulmonary fibrosis, cancers of the respiratory tract, and pleural and peritoneal mesothelioma.

Asbestosis is a diffuse interstitial fibrosing disease of the lung that is directly related to the intensity and duration of exposure. The disease resembles other forms of diffuse interstitial fibrosis (Chap. 293). Usually, exposure has taken place for at least 10 years before the disease becomes manifest. The mechanisms by which asbestos fibers induce lung fibrosis are not completely understood but are known to involve oxidative injury due to the generation of reactive oxygen species by the transition metals on the surface of the fibers as well as from cells engaged in phagocytosis.

Past exposure to asbestos is specifically indicated by pleural plaques on chest radiographs, which are characterized by either thickening or calcification along the parietal pleura, particularly along the lower lung fields, the diaphragm, and the cardiac border. Without additional manifestations, pleural plaques imply only exposure, not pulmonary impairment. Benign pleural effusions also may occur.

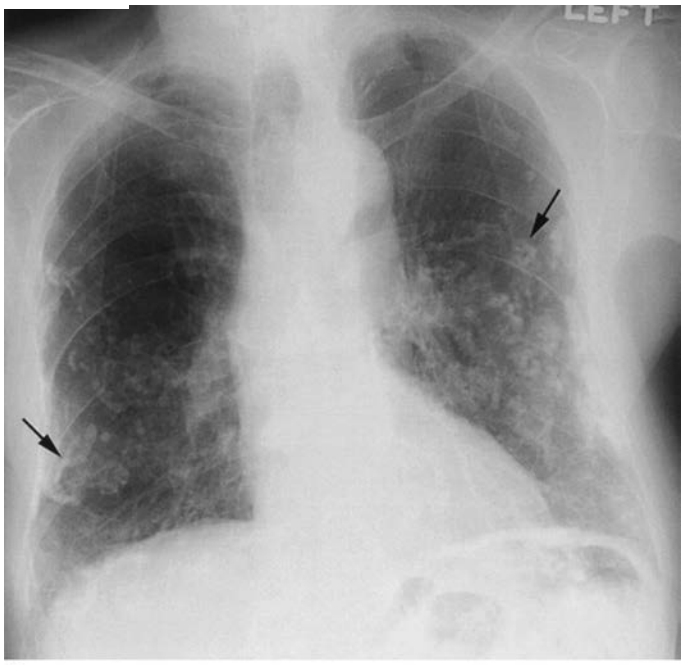
Irregular or linear opacities that usually are first noted in the lower lung fields are the chest radiographic hallmark of asbestosis. An indistinct heart border or a "ground-glass" appearance in the lung fields may be seen. HRCT may show distinct changes of subpleural curvilinear lines 5–10 mm in length that appear to be parallel to the pleural surface (Fig. 289-1).

Pulmonary function testing in asbestosis reveals a restrictive pattern with a decrease in both lung volumes and diffusing capacity. There may also be evidence of mild airflow obstruction (due to peribronchiolar fibrosis).

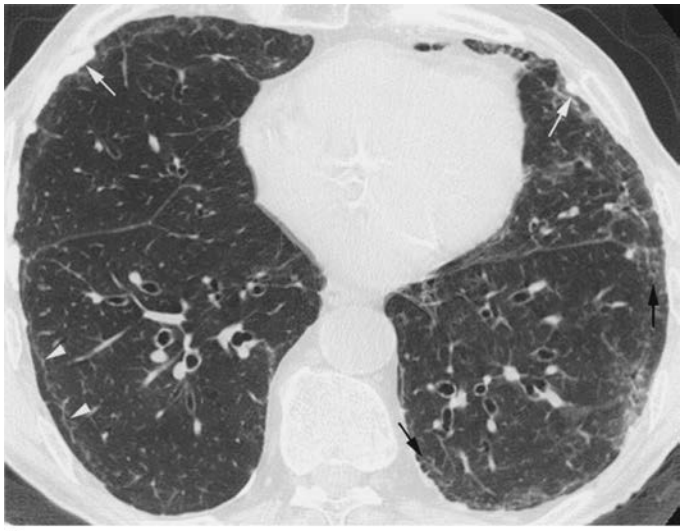
Because no specific therapy is available for asbestosis, supportive care is the same as that given to any patient with diffuse interstitial fibrosis of any cause. In general, newly diagnosed cases will have resulted from exposures that occurred many years before.

Lung cancer (Chap. 78) is the most common cancer associated with asbestos exposure. The excess frequency of lung cancer (all histologic types) in asbestos workers is associated with a minimum latency of 15–19 years between first exposure and development of the disease. Persons with more exposure are at greater risk of disease. In addition, there is a significant interactive effect of smoking and asbestos exposure that results in greater risk than what would be expected from the additive effect of each factor.

Mesotheliomas (Chap. 294), both pleural and peritoneal, are also associated with asbestos exposure. In contrast to lung cancers, these tumors do not appear to be associated with smoking. Relatively short-term asbestos exposures of ≤1–2 years, occurring up to 40 years in the past, have been associated with the development of mesotheliomas (an observation that emphasizes the importance of obtaining a complete environmental exposure history). Although the risk of mesothelioma is much less than that of lung cancer among asbestos-exposed workers, ~3000 cases per year are diagnosed in the United States.



A



B

FIGURE 289-1 Asbestosis. ***A.*** Frontal chest radiograph shows bilateral calcified pleural plaques consistent with asbestos-related pleural disease. Poorly defined linear and reticular abnormalities are seen in the lower lobes bilaterally. ***B.*** Axial high-resolution computed tomography of the thorax obtained through the lung bases shows bilateral, subpleural reticulation (*black arrows*), representing fibrotic lung disease due to asbestosis. Subpleural lines are also present (*arrowheads*), characteristic of, though not specific for, asbestosis. Calcified pleural plaques representing asbestos-related pleural disease (*white arrows*) are also evident.

Because epidemiologic studies have shown that >80% of mesotheliomas may be associated with asbestos exposure, documented mesothelioma in a patient with occupational or environmental exposure to asbestos may be compensable.

SILICOSIS

Despite being one of the oldest known occupational pulmonary hazards, *free silica* (SiO_2), or crystalline quartz, is still a major cause of disease. The major occupational exposures include mining; stonecutting; sand blasting; glass and cement manufacturing; foundry work; packing of silica flour; and quarrying, particularly of granite. Most often, pulmonary fibrosis due to silica exposure (silicosis) occurs in a dose-response fashion after many years of exposure. Two recent outbreaks of silicosis have involved sandblasting of denim jeans to make

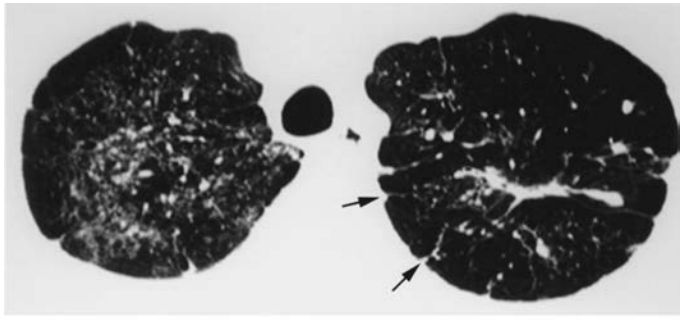
AFKEBOOKS
SINCE 2013



FIGURE 289-2 Acute silicosis. This high-resolution computed tomography scan shows multiple small nodules consistent with silicosis but also diffuse ground-glass densities with thickened intralobular and interlobular septa producing polygonal shapes. This has been referred to as "crazy paving."



A



B

FIGURE 289-3 Chronic silicosis. ***A.*** Frontal chest radiograph in a patient with silicosis shows variably sized, poorly defined nodules (*arrows*) predominating in the upper lobes. ***B.*** Axial thoracic computed tomography image through the lung apices shows numerous small nodules, more pronounced in the right upper lobe. A number of the nodules are subpleural in location (*arrows*).

them look "used," and manufacture and installation of artificial stone ("faux granite") kitchen countertops.

Workers heavily exposed through sandblasting in confined spaces, tunneling through rock with a high quartz content (15–25%), or the manufacture of abrasive soaps may develop acute silicosis with as little as 10 months of exposure. The clinical and pathologic features of acute silicosis are similar to those of pulmonary alveolar proteinosis **(Chap. 293)**. The chest radiograph may show profuse miliary infiltration or consolidation, and there is a characteristic HRCT pattern known as "crazy paving" **(Fig. 289-2)**. The disease may be quite severe and progressive despite the discontinuation of exposure. Whole-lung lavage may provide symptomatic relief and slow the progression.

With long-term, less intense exposure, small rounded opacities in the upper lobes may appear on the chest radiograph after 15–20 years of exposure, usually without associated impairment of lung function (*simple silicosis*). Calcification of hilar nodes may occur in as many as 20% of cases and produces a characteristic "eggshell" pattern. Silicotic nodules may be identified more readily by HRCT **(Fig. 289-3)**. The nodular fibrosis may be progressive in the absence of further exposure, with coalescence and formation of nonsegmental conglomerates of irregular masses >1 cm in diameter (*complicated silicosis*). These masses can become quite large, and when this occurs, the term *progressive massive fibrosis* (PMF) is applied. Significant functional impairment with both restrictive and obstructive components may be associated with PMF.

Because silica causes alveolar macrophage dysfunction, patients with silicosis are at greater risk of acquiring lung infections that involve these cells as a primary defense (*Mycobacterium tuberculosis*, atypical mycobacteria, and fungi). Because of the increased risk of active tuberculosis, the recommended treatment of latent tuberculosis in these patients is longer. Silica has immunoadjuvant properties, and another potential clinical complication of silicosis is autoimmune connective tissue disorders such as rheumatoid arthritis and scleroderma. In addition, there are sufficient epidemiologic data that the International Agency for Research on Cancer lists silica as a probable lung carcinogen.

Other, less hazardous silicates include fuller's earth, kaolin, mica, diatomaceous earths, silica gel, soapstone, carbonate dusts, and cement dusts. The production of fibrosis in workers exposed to these agents is believed to be related either to the free silica content of these dusts or, for substances that contain no free silica, to the potentially large dust loads to which these workers may be exposed. Some silicates, including *talc* and *vermiculite*, may be contaminated with asbestos. Fibrosis of lung or pleura, lung cancer, and mesothelioma have been associated with chronic exposure to talc and vermiculite dusts.

COAL WORKER'S PNEUMOCONIOSIS (CWP)

Occupational exposure to *coal dust* can lead to CWP, which has enormous social, economic, and medical significance in every nation in which coal mining is an important industry. Simple radiographically identified CWP is seen in ~10% of all coal miners and in as many as 50% of anthracite miners with >20 years of work on the coal face. The prevalence of disease is lower in workers in bituminous coal mines.

With prolonged exposure to coal dust (i.e., 15–20 years), small, rounded opacities similar to those of silicosis may develop. As in silicosis, the presence of these nodules (*simple CWP*) usually is not associated with pulmonary impairment. In addition to CWP, coal dust can cause chronic bronchitis and COPD **(Chap. 292)**. The effects of coal dust are additive to those of cigarette smoking.

Complicated CWP is manifested by the appearance on the chest radiograph of nodules ≥1 cm in diameter generally confined to the upper half of the lungs. As in silicosis, this condition can progress to PMF that is accompanied by severe lung function deficits and associated with premature mortality. Despite improvements in technology to protect coal miners, cases of PMF still occur in the United States at a disturbing rate.

AFKEBOOKS
SINCE 2013

Caplan syndrome (Chap. 358), first described in coal miners but subsequently in patients with silicosis, is the combination of pneumoconiotic nodules and seropositive rheumatoid arthritis. Silica is often present in anthracitic coal dust, and its presence may contribute to risk of PMF.

CHRONIC BERYLLIUM DISEASE

Beryllium is a lightweight metal with tensile strength, good electrical conductivity, and value in the control of nuclear reactions through its ability to quench neutrons. Although beryllium may produce an acute pneumonitis, it is far more commonly associated with a chronic granulomatous inflammatory disease that is similar to sarcoidosis (Chap. 367). Unless one inquires specifically about occupational exposures to beryllium in the manufacture of alloys, ceramics, or high-technology electronics in a patient with sarcoidosis, one may miss entirely the etiologic relationship to the occupational exposure. What distinguishes CBD from sarcoidosis is evidence of a specific cell-mediated immune response (i.e., delayed hypersensitivity) to beryllium.

The test that usually provides this evidence is the beryllium lymphocyte proliferation test (BeLPT). The BeLPT compares the in vitro proliferation of lymphocytes from blood or bronchoalveolar lavage in the presence of beryllium salts with that of unstimulated cells. Proliferation is usually measured by lymphocyte uptake of radiolabeled thymidine.

Chest imaging findings are similar to those of sarcoidosis (nodules along septal lines) except that hilar adenopathy is somewhat less common. As with sarcoidosis, pulmonary function test results may show restrictive and/or obstructive ventilatory deficits and decreased diffusing capacity. With early disease, both chest imaging studies and pulmonary function tests may be normal. Fiberoptic bronchoscopy with transbronchial lung biopsy usually is required to make the diagnosis of CBD. In a beryllium-sensitized individual, the presence of noncaseating granulomas or monocytic infiltration in lung tissue establishes the diagnosis. Accumulation of beryllium-specific CD4+ T cells occurs in the granulomatous inflammation seen on lung biopsy. Susceptibility to CBD is highly associated with human leukocyte antigen DP (HLA-DP) alleles that have a glutamic acid in position 69 of the β chain.

OTHER METALS

Aluminum and titanium dioxide have been rarely associated with a sarcoid-like reaction in lung tissue. Exposure to dust containing tungsten carbide, also known as "hard metal," may produce giant cell interstitial pneumonitis. Cobalt is a constituent of tungsten carbide and is the likely etiologic agent of both the interstitial pneumonitis and the occupational asthma that may occur. The most common exposures to tungsten carbide occur in tool and dye, saw blade, and drill bit manufacture. Diamond polishing may also involve exposure to cobalt dust. In patients with interstitial lung disease, one should always inquire about exposure to metal fumes and/or dusts. Especially when sarcoidosis appears to be the diagnosis, one should always consider possible CBD.

OTHER INORGANIC DUSTS

Most of the inorganic dusts discussed thus far are associated with the production of either dust macules or interstitial fibrotic changes in the lung. Other inorganic and organic dusts (see categories in Table 289-1), along with some of the dusts previously discussed, are associated with chronic mucus hypersecretion (chronic bronchitis), with or without reduction of expiratory flow rates. Cigarette smoking is the major cause of these conditions, and any effort to attribute some component of the disease to occupational and environmental exposures must take cigarette smoking into account. Most studies suggest an additive effect of dust exposure and smoking. The pattern of the irritant dust effect is similar to that of cigarette smoking, suggesting that small airway inflammation may be the initial site of pathologic response in those cases and continued exposure may lead to chronic bronchitis and COPD.

ORGANIC DUSTS

Some of the specific diseases associated with organic dusts are discussed in detail in the chapters on asthma (Chap. 287) and hypersensitivity pneumonitis (Chap. 288). Many of these diseases are named for the specific setting in which they are found, e.g., farmer's lung, malt worker's disease, and mushroom worker's disease. Often the temporal relation of symptoms to exposure furnishes the best evidence for the diagnosis. Three occupational exposures are singled out for discussion here because they affect the largest proportions of workers.

Cotton Dust (Byssinosis) Workers occupationally exposed to cotton dust (but also to flax, hemp, or jute dust) in the production of yarns for textiles and rope making are at risk for an asthma-like syndrome known as byssinosis. The risk of byssinosis is associated with both cotton dust and endotoxin levels in the workplace environment.

Byssinosis is characterized clinically as occasional (early-stage) and then regular (late-stage) chest tightness toward the end of the first day of the workweek ("Monday chest tightness"). Exposed workers may show a significant drop in FEV_1 over the course of a Monday workshift. Initially the symptoms do not recur on subsequent days of the week, but in a subset of workers, chest tightness may recur or persist throughout the workweek. After >10 years of exposure, workers with recurrent symptoms are more likely to have an obstructive pattern on pulmonary function testing.

Dust exposure can be reduced by the use of exhaust hoods, general increases in ventilation, and wetting procedures, but respiratory protective equipment may be required during certain operations. Regular surveillance of pulmonary function in cotton dust–exposed workers using spirometry before and after the workshift is required by OSHA. All workers with persistent symptoms or significantly reduced levels of pulmonary function should be moved to areas of lower risk of exposure.

Grain Dust Worldwide, many farmers and workers in grain storage facilities are exposed to grain dust. The presentation of obstructive airway disease in grain dust–exposed workers is virtually identical to the characteristic findings in cigarette smokers, i.e., persistent cough, mucus hypersecretion, wheeze and dyspnea on exertion, and reduced FEV_1 and FEV_1/FVC (forced vital capacity) ratio (Chap. 285).

Dust concentrations in grain elevators vary greatly but can be >10,000 μg/m^3 with many particles in the respirable size range. The effect of grain dust exposure is additive to that of cigarette smoking, with ~50% of workers who smoke having symptoms. Smoking grain dust–exposed workers are more likely to have obstructive ventilatory deficits on pulmonary function testing. As in byssinosis, endotoxin may play a role in grain dust–induced chronic bronchitis and COPD.

Farmer's Lung This condition results from exposure to moldy hay containing spores of thermophilic actinomycetes that produce a hypersensitivity pneumonitis (Chap. 288). A patient with acute farmer's lung presents 4–8 h after exposure with fever, chills, malaise, cough, and dyspnea without wheezing. The history of exposure is obviously essential to distinguish this disease from influenza or pneumonia with similar symptoms. In the chronic form of the disease, the history of repeated attacks after similar exposure is important in differentiating this syndrome from other causes of patchy fibrosis (e.g., sarcoidosis).

A wide variety of other organic dusts are associated with the occurrence of hypersensitivity pneumonitis (Chap. 288). For patients who present with hypersensitivity pneumonitis, specific and careful inquiry about occupations, hobbies, and other home environmental exposures is necessary to uncover the source of the etiologic agent.

TOXIC CHEMICALS

Exposure to toxic chemicals affecting the lung generally involves gases and vapors. A common accident is one in which the victim is trapped in a confined space where the chemicals have accumulated to harmful levels. In addition to the specific toxic effects of the chemical, the victim often sustains considerable anoxia, which can play a dominant role in determining whether the individual survives.

Table 289-2 lists a variety of toxic agents that can produce acute and sometimes life-threatening reactions in the lung. All these agents in sufficient concentrations have been demonstrated, at least in animal studies, to affect the lower airways and disrupt alveolar architecture, either acutely or as a result of chronic exposure.

AFKEBOOKS
SINCE 2013

TABLE 289-2 Selected Common Toxic Chemical Agents That Affect the Lung

AGENT(S)	SELECTED EXPOSURES	ACUTE EFFECTS FROM HIGH OR ACCIDENTAL EXPOSURE	CHRONIC EFFECTS FROM RELATIVELY LOW EXPOSURE
Acid anhydrides	Manufacture of resin esters, polyester resins, thermoactivated adhesives	Nasal irritation, cough	Asthma, chronic bronchitis, hypersensitivity pneumonitis
Acid fumes: H_2SO_4, HNO_3	Manufacture of fertilizers, chlorinated organic compounds, dyes, explosives, rubber products, metal etching, plastics	Mucous membrane irritation, followed by chemical pneumonitis 2–3 days later	Bronchitis and suggestion of mildly reduced pulmonary function in children with lifelong residential exposure to high levels
Acrolein and other aldehydes	By-product of burning plastics, woods, tobacco smoke	Mucous membrane irritant, decrease in lung function	Upper respiratory tract irritation
Ammonia	Refrigeration; petroleum refining; manufacture of fertilizers, explosives, plastics, and other chemicals	Same as for acid fumes, but bronchiectasis also has been reported	Upper respiratory tract irritation, chronic bronchitis
Cadmium fumes	Smelting, soldering, battery production	Mucous membrane irritant, acute respiratory distress syndrome (ARDS)	Chronic obstructive pulmonary disease (COPD)
Formaldehyde	Manufacture of resins, leathers, rubber, metals, and woods; laboratory workers, embalmers; emission from urethane foam insulation	Same as for acid fumes	Nasopharyngeal cancer
Halides and acid salts (Cl, Br, F)	Bleaching in pulp, paper, textile industry; manufacture of chemical compounds; synthetic rubber, plastics, disinfectant, rocket fuel, gasoline	Mucous membrane irritation, pulmonary edema; possible reduced forced vital capacity (FVC) 1–2 years after exposure	Upper respiratory tract irritation, epistaxis, tracheobronchitis
Hydrogen sulfide	By-product of many industrial processes, oil, other petroleum processes and storage	Increase in respiratory rate followed by respiratory arrest, lactic acidosis, pulmonary edema, death	Conjunctival irritation, chronic bronchitis, recurrent pneumonitis
Isocyanates (TDI, HDI, MDI)	Production of polyurethane foams, plastics, adhesives, surface coatings	Mucous membrane irritation, dyspnea, cough, wheeze, pulmonary edema	Upper respiratory tract irritation, cough, asthma, hypersensitivity pneumonitis, reduced lung function
Nitrogen dioxide	Silage, metal etching, explosives, rocket fuels, welding, by-product of burning fossil fuels	Cough, dyspnea, pulmonary edema may be delayed 4–12 h; possible result from acute exposure: bronchiolitis obliterans in 2–6 weeks	Emphysema in animals, chronic bronchitis, associated with reduced lung function growth in children with lifelong residential exposure
Ozone	Arc welding, flour bleaching, deodorizing, emissions from copying equipment, photochemical air pollutant	Mucous membrane irritant, reduced pulmonary function transiently in children and adults, asthma exacerbation	Excess cardiopulmonary mortality rates, increased risk for new-onset asthma in children
Phosgene	Organic compound, metallurgy, volatilization of chlorine-containing compounds	Delayed onset of bronchiolitis and pulmonary edema	Chronic bronchitis
Sulfur dioxide	Manufacture of sulfuric acid, bleaches, coating of nonferrous metals, food processing, refrigerant, burning of fossil fuels, wood pulp industry	Mucous membrane irritant, epistaxis, bronchospasm (especially in people with asthma)	Chronic bronchitis

Abbreviations: HDI, hexamethylene diisocyanate; MDI, methylene diphenyl diisocyanate; TDI, toluene diisocyanate.

Firefighters and fire victims are at risk of *smoke inhalation*, an important cause of acute cardiorespiratory failure. Smoke inhalation kills more fire victims than does thermal injury. Carbon monoxide poisoning with resulting significant hypoxemia can be life-threatening (Chap. 459). Synthetic materials (plastic, polyurethanes), when burned, may release a variety of other toxic agents (such as cyanide and hydrochloric acid), and this must be considered in evaluating smoke inhalation victims. Exposed victims may have some degree of lower respiratory tract inflammation and/or pulmonary edema.

Exposure to certain highly reactive, low-molecular-weight agents used in the manufacture of synthetic polymers, paints, and coatings (*diisocyanates* in polyurethanes, *aromatic amines* and *acid anhydrides* in epoxies) is associated with a high risk of occupational asthma. Although this occupational asthma manifests clinically as if sensitization has occurred, an IgE antibody–mediated mechanism is not necessarily involved. Hypersensitivity pneumonitis–like reactions also have been described in diisocyanate and acid anhydride–exposed workers.

Fluoropolymers such as Teflon, which at normal temperatures produce no reaction, become volatilized upon heating. The inhaled agents cause a characteristic syndrome of fever, chills, malaise, and occasionally mild wheezing, leading to the diagnosis of *polymer fume fever*. A similar self-limited, influenza-like syndrome—*metal fume fever*—results from acute exposure to fumes containing zinc oxide, typically from welding of galvanized steel. These inhalational fever syndromes may begin several hours after work and resolve within 24 h, only to return on repeated exposure.

Two other agents have been associated with potentially severe lung disease. Occupational exposure to nylon flock has been shown to induce a lymphocytic bronchiolitis, and workers exposed to diacetyl, which is used to provide "butter" flavor in the manufacture of microwave popcorn and other foods, have developed bronchiolitis obliterans (Chap. 293).

World Trade Center Disaster A consequence of the attack on the World Trade Center (WTC) on September 11, 2001, was relatively heavy exposure of a large number of firefighters and other rescue workers to the dust generated by the collapse of the buildings. Environmental monitoring and chemical characterization of WTC dust have revealed a wide variety of potentially toxic constituents, although much of the dust was pulverized cement. Possibly because of the high alkalinity of WTC dust, significant cough, wheeze, and phlegm production occurred among firefighters and cleanup crews. New cough and wheeze syndromes also occurred among local residents. Heavier exposure to WTC dust among New York City firefighters was associated with accelerated decline of lung function over the first year after the disaster. More recently, concerns have been raised about risk of interstitial lung disease, especially of a granulomatous nature.

OCCUPATIONAL RESPIRATORY CARCINOGENS

Exposures at work have been estimated to contribute to 10% of all lung cancer cases. In addition to asbestos, other agents either proven or suspected to be respiratory carcinogens include acrylonitrile, arsenic

AFKEBOOKS
SINCE 2013

compounds, beryllium, bis(chloromethyl) ether, chromium (hexavalent), formaldehyde (nasal), isopropanol (nasal sinuses), mustard gas, nickel carbonyl (nickel smelting), polycyclic aromatic hydrocarbons (coke oven emissions and diesel exhaust), secondhand tobacco smoke, silica (both mining and processing), talc (possible asbestos contamination in both mining and milling), vinyl chloride (sarcomas), wood (nasal), and uranium. Workers at risk of radiation-related lung cancer include not only those involved in mining or processing uranium but also those exposed in underground mining operations of other ores where radon daughters may be emitted from rock formations.

ASSESSMENT OF DISABILITY

Disability is the term used to describe the decreased ability to work due to the effects of a medical condition. Physicians are generally able to assess physiologic dysfunction, or *impairment*, but the rating of disability for compensation of loss of income also involves nonmedical factors such as the education and employability of the individual. The disability rating scheme differs with the compensation-granting agency. For example, the U.S. Social Security Administration requires that an individual be unable to do any work (i.e., *total* disability) before he or she will receive income replacement payments. Many state workers' compensation systems allow for payments for *partial* disability. In the Social Security scheme, no determination of cause is done, whereas work-relatedness must be established in workers' compensation systems.

For respiratory impairment rating, resting pulmonary function tests (spirometry and diffusing capacity) are used as the initial assessment tool, with cardiopulmonary exercise testing (to assess maximal oxygen consumption) used if the results of the resting tests do not correlate with the patient's symptoms. Methacholine challenge (to assess airway reactivity) can also be useful in patients with asthma who have normal spirometry when evaluated. Some compensation agencies (e.g., Social Security) have proscribed disability classification schemes based on pulmonary function test results. When no specific scheme is proscribed, the *Guidelines of the American Medical Association* should be used.

GENERAL ENVIRONMENTAL EXPOSURES

OUTDOOR AIR POLLUTION

Primary standards regulated by the U.S. Environmental Protection Agency (EPA) designed to protect the public health with an adequate margin of safety exist for sulfur dioxide, particulate matter (PM), nitrogen dioxide, ozone, lead, and carbon monoxide. Standards for each of these pollutants are updated regularly through an extensive review process conducted by the EPA. (For details on current standards, go to *https://www.epa.gov/criteria-air-pollutants/naaqs-table*).

Pollutants are generated from both stationary sources (power plants and industrial facilities) and mobile sources (motor vehicles), and none of the regulated pollutants occurs in isolation. Furthermore, pollutants may be changed by chemical reactions after being emitted. For example, sulfur dioxide and PM emissions from a coal-fired power plant may react in air to produce acid sulfate aerosols, which can be transported long distances in the atmosphere. Oxides of nitrogen and volatile organic compounds from automobile exhaust react with sunlight to produce ozone. Although originally recognized in Los Angeles, photochemically derived pollution ("smog") is now known to be a problem throughout the United States and in many other countries. Both acute and chronic effects of pollutant exposures have been documented in large population studies.

The symptoms and diseases associated with air pollution are the same as conditions commonly associated with cigarette smoking. In addition, decreased growth of lung function and asthma have been associated with chronic exposure to only modestly elevated levels of traffic-related air pollution. Multiple population-based time-series studies within cities have demonstrated excess health care utilization for asthma and other cardiopulmonary conditions as well as increased mortality rates. Cohort studies comparing cities that have relatively high levels of particulate exposures with less polluted communities suggest excess morbidity and mortality rates from cardiopulmonary conditions in long-term residents of the former. The strong epidemiologic evidence that fine PM is a risk factor for cardiovascular morbidity and mortality has prompted toxicologic investigations into the underlying mechanisms. The inhalation of fine particles from combustion sources generates oxidative stress followed by local injury and inflammation in the lungs that in turn lead to autonomic and systemic inflammatory responses. Recent research findings on the health effects of air pollutants have led to stricter U.S. ambient air quality standards for ozone, oxides of nitrogen, and PM as well as greater emphasis on publicizing pollution alerts to encourage individuals with cardiovascular and respiratory disorders to stay indoors during high-pollution episodes (e.g., from wildfires).

INDOOR EXPOSURES

Secondhand tobacco smoke (Chap. 454), radon gas, wood smoke, and other biologic agents generated indoors must be considered. Several studies have shown that the respirable particulate load in any household is directly proportional to the number of cigarette smokers living in that home. Increases in prevalence of respiratory illnesses, especially asthma, and reduced levels of pulmonary function have been found in the children of smoking parents in a number of studies. Recent meta-analyses for lung cancer and cardiopulmonary diseases, combining data from multiple secondhand tobacco smoke epidemiologic studies, suggest an ~25% increase in relative risk for each condition, even after adjustment for major potential confounders.

Exposure to *radon gas* in homes is a risk factor for lung cancer. The main radon product (radon-222) is a gas that results from the decay series of uranium-238, with the immediate precursor being radium-226. The amount of radium in earth materials determines how much radon gas will be emitted. Levels associated with excess lung cancer risk may be present in as many as 10% of the houses in the United States. When smokers reside in the home, the problem is potentially greater, because the molecular size of radon particles allows them to attach readily to smoke particles that are inhaled. Fortunately, technology is available for assessing and reducing the level of exposure.

Other indoor exposures of concern are bioaerosols that contain antigenic material (fungi, cockroaches, dust mites, and pet danders) associated with an increased risk of atopy and asthma. Indoor chemical agents include strong cleaning agents (bleach, ammonia), formaldehyde, perfumes, pesticides, and oxides of nitrogen from gas appliances. Nonspecific responses associated with "tight-building syndrome," perhaps better termed "building-associated illness," in which no particular agent has been implicated, have included a wide variety of complaints, among them respiratory symptoms that are relieved only by avoiding exposure in the building in question. The degree to which "smells" and other sensory stimuli are involved in the triggering of potentially incapacitating psychological or physical responses has yet to be determined, and the long-term consequences of such environmental exposures are unknown.

Indoor exposure to *household air pollution* from cooking or heating with solid fuels (wood, dung, crop residues, charcoal, coal) is estimated to be responsible for ~2.7% of worldwide disability-adjusted life-years (DALYs) lost, due to acute lower respiratory infections in children, COPD and lung cancer in women, and cardiovascular disease among men. This burden of disease places exposure to household air pollution as one of the leading environmental hazards for poor health on a global scale.

Forty percent of the world's population uses solid fuel for cooking, heating, or baking. Kerosene (similar to diesel fuel) is often used for lighting and sometimes cooking. This occurs predominantly in the rural areas of developing countries. Because many families burn coal or biomass fuels in open stoves, which are highly inefficient, and inside homes with poor ventilation, women and young children are exposed on a daily basis to high levels of smoke. In these homes, 24-h mean levels of fine PM have been reported to be 2–30 times higher than the National Ambient Air Quality Standard set by the U.S. EPA.

Epidemiologic studies have consistently shown associations between exposure to biomass smoke and both chronic bronchitis and COPD. Because of increased migration to the United States from developing countries, clinicians need to be aware of the chronic respiratory effects

AFKEBOOKS
SINCE 2013

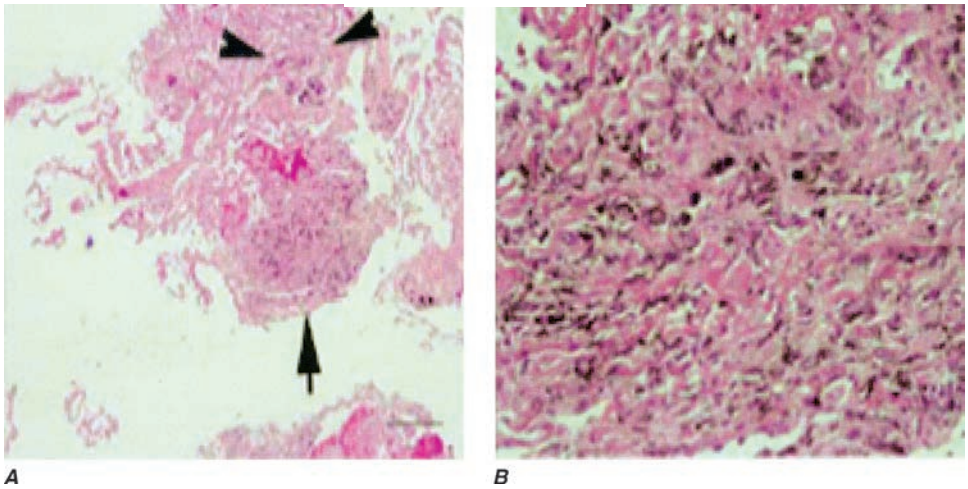


FIGURE 289-4 Histopathologic features of biomass smoke–induced interstitial lung disease. ***A.*** Anthracitic pigment is seen accumulating along alveolar septae (*arrowheads*) and within a pigmented dust macule (*single arrow*). ***B.*** A high-power photomicrograph contains a mixture of fibroblasts and carbon-laden macrophages.

of exposure to biomass smoke, which can include interstitial lung disease (Fig. 289-4). Evidence is beginning to emerge that improved stoves that reduce biomass smoke exposure can reduce risk of respiratory illness in both children and adults.

Household air pollution (HAP) from domestic use of solid fuels also contributes substantially to outdoor air pollution. Contributions from HAP, coal-fired power plants without emission scrubbers, and increased traffic congestion involving motor vehicles without pollution controls can lead to high concentrations of outdoor air pollution, especially fine PM, in mega-cities in developing countries (e.g., Delhi).

Acknowledgment

The author acknowledges the contribution of Dr. Frank Speizer to the prior version of this chapter.

FURTHER READING

Almberg KS et al: Progressive massive fibrosis resurgence identified in U.S. coal miners filing for Black Lung Benefits, 1970-2016. Ann Am Thorac Soc 15:1420, 2018.

Balmes JR: Household air pollution from domestic combustion of solid fuels and health. J Allergy Clin Immunol 143:1979, 2019.

Blanc PD et al: The occupational burden of nonmalignant respiratory diseases. An official American Thoracic Society and European Respiratory Society statement. Am J Respir Crit Care Med 199:1312, 2019.

Gauderman WJ et al: Association of improved air quality with lung development in children. N Engl J Med 372:905, 2015.

Leon-Jimenez A et al: Artificial stone silicosis: Rapid progression following exposure cessation. Chest 158:1060, 2020.

Musk AW et al: Asbestos-related diseases. Int J Tuberc Lung Dis 24:562, 2020.

290 Bronchiectasis

Rebecca M. Baron, Beverly W. Baron, Miriam Baron Barshak

Bronchiectasis refers to an irreversible airway dilation that involves the lung in either a focal or a diffuse manner and that classically has been categorized as cylindrical or tubular (the most common form), varicose, or cystic. This chapter will focus largely on non–cystic fibrosis (CF) bronchiectasis. **The reader is referred to Chapter 291 for a more focused discussion on CF bronchiectasis.**

ETIOLOGY

Bronchiectasis can arise from infectious or noninfectious causes (Table 290-1). Clues to the underlying etiology often are provided by the pattern of lung involvement. *Focal bronchiectasis* refers to bronchiectatic changes in a localized area of the lung and can be a consequence of obstruction of the airway—either extrinsic (e.g., due to compression by adjacent lymphadenopathy or parenchymal tumor mass) or intrinsic (e.g., due to an airway tumor or aspirated foreign body, a scarred/stenotic airway, or bronchial atresia from congenital underdevelopment of the airway). *Diffuse bronchiectasis* is characterized by widespread bronchiectatic changes throughout the lung and often arises from an underlying systemic or infectious disease process.

More pronounced involvement of the upper lung fields is most common in CF and also is observed in postradiation fibrosis, corresponding to the lung region encompassed by the radiation port. Bronchiectasis with predominant involvement of the lower lung fields usually has its source in chronic recurrent aspiration (e.g., due to esophageal motility disorders like those in scleroderma), end-stage fibrotic lung disease (e.g., traction bronchiectasis from idiopathic pulmonary fibrosis), or recurrent immunodeficiency-associated infections (e.g., hypogammaglobulinemia). Bronchiectasis resulting from infection by nontuberculous mycobacteria (NTM), most commonly the *Mycobacterium avium-intracellulare* complex (MAC), often preferentially affects the midlung fields. Congenital causes of bronchiectasis with predominant midlung field involvement include the dyskinetic/immotile cilia syndrome. Finally, predominant involvement of the central airways is reported in association with allergic bronchopulmonary aspergillosis (ABPA), in which an immune-mediated reaction to *Aspergillus* damages the bronchial wall. Congenital causes of central airway–predominant bronchiectasis resulting from cartilage deficiency include tracheobronchomegaly (Mounier-Kuhn syndrome) and Williams-Campbell syndrome.

In many cases, the etiology of bronchiectasis is not determined. In case series, as many as 25–50% of patients referred for bronchiectasis have idiopathic disease.

EPIDEMIOLOGY

The overall reported prevalence of bronchiectasis in the United States has recently increased, but the epidemiology of bronchiectasis varies greatly with the underlying etiology. For example, patients born with CF often develop significant clinical bronchiectasis in late adolescence or early adulthood, although atypical presentations of CF in adults in their thirties and forties are also possible. In contrast, bronchiectasis resulting from MAC infection classically affects nonsmoking women >50 years of age. In general, the incidence of bronchiectasis increases with age. Bronchiectasis is more common among women than among men. Bronchiectasis may also frequently be co-diagnosed with chronic obstructive pulmonary disease (COPD) or asthma.

AFKEBOOKS
SINCE 2013

293 Interstitial Lung Disease

Gary M. Hunninghake, Ivan O. Rosas

Diffuse parenchymal lung diseases include a large number (>200) of heterogeneous conditions that affect the lung parenchyma with varying degrees of inflammation and fibrosis. While remodeling of the interstitial space, the region between the epithelium and endothelium, tends to be the dominant site of involvement for most of the interstitial lung diseases (ILDs), it is important to recognize the prominent role of the alveolar epithelium and endothelial cells (including both airways and vessels) in the pathogenesis of these ILDs.

Despite the diverse array of conditions, most patients ultimately diagnosed with an ILD will come to medical attention with reports of progressive exertional dyspnea or a persistent dry cough. However, because some ILDs are part of multisystem disorders, some patients will be identified based on nonrespiratory symptomatology (e.g., skin thickening in the setting of systemic sclerosis, **Chap. 359**) or physical examination findings (e.g., ulnar deviation of the fingers in the setting of rheumatoid arthritis [RA], **Chap. 358**). Additionally, ILDs can also be identified incidentally based on the results of abnormal pulmonary function tests, chest x-rays (CXRs), computed tomography (CT) studies of both the chest and abdomen (which can both visualize, at least a portion, of the lung parenchyma), and positron emission tomography (PET) scans. It is important to remember that ILDs can be associated with high rates of morbidity and mortality, and although prognosis depends on both disease extent and specificity, this fact makes these important disorders to recognize in a timely manner.

Owing to a variety of clinical presentations, as well as overlapping imaging and histopathologic findings **(Table 293-1)**, ILDs can be difficult to diagnose. A generally accepted central tenet of ILD diagnosis is that the combined weight of clinical data, laboratory studies, pulmonary function testing, imaging findings, and histopathology (if obtained) are jointly required to make a confident diagnosis. No single piece of data confers a diagnosis alone. For example, a lung biopsy demonstrating a usual interstitial pneumonia (UIP) pattern is helpful in diagnosing a patient with idiopathic pulmonary fibrosis (IPF) but can also be present in some connective tissue diseases (CTDs) (e.g., RA-associated ILD, **Chap. 358**). In light of this challenge, most ILD centers recommend a multidisciplinary approach to the diagnosis (and in some cases the management) of ILDs. An example of a multidisciplinary approach might include a conference attended by pulmonologists, rheumatologists, radiologists, and pathologists where all of the data generated on a patient can be discussed and reviewed jointly by those with unique sets of expertise in the care of patients with ILD.

While there are numerous ways to categorize the ILDs, one classic approach is to divide the ILDs into those of known and unknown causes **(Fig. 293-1)**. Although even this approach has limitations (e.g., genetic studies demonstrate that a significant portion of familial pulmonary fibrosis and IPF [classically described as diseases of unknown cause] may be explained, in part, by genetic factors), it is a useful place to start. Known causes of ILD include occupational exposures (e.g., asbestosis), medications (e.g., nitrofurantoin), and those related to an underlying systemic disease (e.g., cryptogenic organizing pneumonia [COP] in the setting of polymyositis). Unknown causes of ILD include groups of rare disorders often with classic presentations (e.g., a spontaneous pneumothorax in a young female with diffuse cystic changes on a chest CT might suggest lymphangioleiomyomatosis [LAM]) and the most common group of ILDs, the idiopathic interstitial pneumonias (IIPs). Granulomatous lung diseases straddle both known (e.g., hypersensitivity pneumonitis [HP] due to chronic bird exposure, **Chap. 288**) and unknown (e.g., sarcoidosis, **Chap. 367**) causes and are often separated due to their unique presentations, imaging findings, and diagnostic evaluation. Equally important to knowledge of disease classification is knowledge of disease prevalence. Although there is variability within different demographic groups, most studies demonstrate that IPF,

TABLE 293-1 Common Interstitial Lung Disease (ILD) Findings

	IPF	NONSPECIFIC INTERSTITIAL PNEUMONIA	RESPIRATORY BRONCHIOLITIS ASSOCIATED ILD	SYSTEMIC SCLEROSIS ASSOCIATED ILD	SARCOIDOSIS
Clinical symptoms	Gradual onset of SOB, dry cough. Unusual in older adults.	Subacute onset of SOB, dry cough. Frequently associated with other conditions.	Can be asymptomatic, or have SOB and cough.	Gradual onset of SOB, dry cough. Fatigue, tightening of skin, exaggerated cold response, reflux, and difficulty swallowing.	Can be asymptomatic, or have SOB and cough. Can also have fatigue, palpitations, eye, skin, and joint findings.
Physical exam findings	Frequent rales at lung bases; digital clubbing is common.	Frequent rales. Clubbing is less common.	Rales common. Clubbing is rare.	Can have rales in isolation. Also skin thickening, joint swelling, and telangiectasias.	Can be normal; rales may be present. Can have skin findings, joint pain, and enlarged lymph nodes.
Exposures	Idiopathic but many exposed to smoke. Genetic findings may explain more than one-third of the risk of the disease.	Can be idiopathic but should prompt consideration for associated conditions.	Strong association with smoking.	Mostly unknown; some debate about solvent and silicate exposures.	Mostly unknown, although silicate dusts thought to play a role in some cases.
HRCT findings	Bilateral subpleural reticular changes most prominent in lower, posterior lung zones. Traction bronchiectasis and honeycombing common. Classic usual interstitial pneumonia (UIP) pattern is considered diagnostic.	Peripheral subpleural ground glass and reticular patterns. Traction bronchiectasis is common, but honeycombing is rare. HRCT not diagnostic.	Diffuse patchy centrilobular ground glass nodules.	Can have UIP or nonspecific interstitial pneumonia (NSIP) patterns, also dilated esophagus, occasional mediastinal calcifications, and pulmonary vascular enlargement.	Can have mediastinal and hilar lymphadenopathy. Peribronchovascular reticular-nodular findings.
Histopathology	UIP pattern including fibroblastic foci, temporal and spatial heterogeneity, honeycombing.	Cellular or fibrotic pattern of NSIP. More uniform than a UIP pattern.	Respiratory bronchiolitis with adjacent inflammatory and fibrosing changes. Pigment-laden macrophages.	Both UIP or NSIP patterns can occur.	Noncaseating granulomas.
Clinical course	50% 3- to 5-year mortality.	18% 5-year mortality.	25% 7-year mortality.	20–30% 10-year mortality.	Generally low but varies by state.

Abbreviations: HRCT, high-resolution computed tomography; IPF, idiopathic pulmonary fibrosis; SOB, shortness of breath.

AFKEBOOKS
SINCE 2013

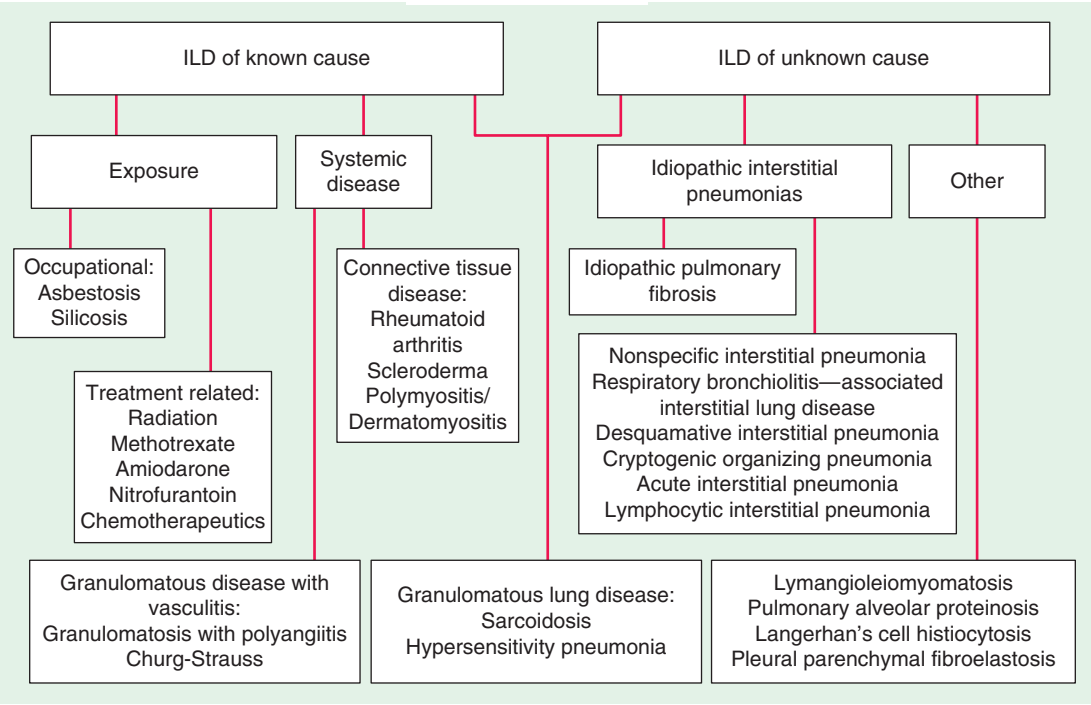


FIGURE 293-1 Classification of interstitial lung disease. This algorithm represents a common approach to subclassifying the interstitial lung diseases. It is typical to divide the interstitial lung diseases into those of known and unknown causes (although it is important to note that genetic studies demonstrate that a significant portion of familial and idiopathic pulmonary fibrosis [classically described as diseases of unknown cause] may be explained, in part, by genetic factors). The idiopathic interstitial pneumonias were more precisely defined by a 2002 study as described in Am J Respir Crit Care Med 165:277, 2002, referenced in the Further Reading list.

sarcoidosis **(Chap. 367)**, and ILDs related to CTDs **(Chap. 413)** as a group are among the most common forms of ILD.

DIAGNOSTIC APPROACH

The initial diagnostic approach to diffuse parenchymal lung disease is often broader than a focus on ILD and should include an evaluation for alternate causes including cardiovascular disease (e.g., heart failure, **Chap. 258**), diffuse infections (e.g., *pneumocystis* pneumonia, **Chap. 220**), and malignancy (e.g., bronchoalveolar cell carcinoma, **Chap. 315 in HPIM 19e**). This chapter will focus on the diagnostic evaluation that helps to distinguish among the various forms of ILD.

HISTORY

Age Age at presentation has a strong influence on the pretest probability that IPF, in particular, is present. For example, IPF occurs most commonly in patients aged >60 and is quite rare among patients aged <50. In fact, in patients aged >65 without strong evidence for an alternate diagnosis, atypical chest CT findings are still more likely to result in a histopathologic diagnosis of UIP (a pathologic hallmark of IPF) than they are to result in an alternate IIP diagnosis. Other common ILDs, such as sarcoidosis and CTD-associated ILD, and less common ILDs, such as LAM and pulmonary Langerhans cell histiocytosis (PLCH), tend to present between the ages of 20 and 40.

Sex Although less influential than age, sex has some influence on likelihood of various ILDs. LAM (and the related disorder tuberous sclerosis) **(see Chap. 315 in HPIM 19e)** is a disorder that is frequently diagnosed in young women. Many CTD-associated ILDs are more common among women, with the exception of RA-associated ILD, which is more common among men. IPF and occupational/exposure-related ILDs (likely due to work-related exposures that tend to differ between men and women) are more common among men.

Duration of Symptoms Acute presentations (*days to weeks*) of ILD are unusual and are commonly misdiagnosed as more common diseases such as pneumonia, a chronic obstructive pulmonary disease (COPD) exacerbation, or heart failure. ILDs that can present acutely include eosinophilic pneumonia, acute interstitial pneumonia (AIP), HP, and granulomatosis with polyangiitis (GPA). An acute exacerbation of IPF as the initial presentation of this disease should also be a consideration given its prevalence. ILDs most commonly have a chronic indolent presentation (*months to years*) typified by IPF. However, subacute presentations (*weeks to months*) can occur in most of the ILDs, but in the right context could suggest sarcoidosis, CTD-associated ILD, drug-induced ILD, or COP.

Respiratory Symptoms Progressive dyspnea, most frequently noted with exertion, is the most common complaint in patients presenting with an ILD. Despite this fact, both research studies of general population samples and clinical experiences of asymptomatic patient referrals with abnormal chest CT imaging patterns have also demonstrated that some patients, even those with more extensive disease, may not report dyspnea. Cough, particularly a dry cough, is also common and can be the most prominent symptom in patients with IPF. Cough is often reported in other ILDs, particularly those that have prominent airway involvement including sarcoidosis and HP. Cough with hemoptysis is rare and could suggest an ILD associated with diffuse alveolar hemorrhage (DAH) (e.g., Goodpasture's syndrome), GPA, or LAM. Cough with hemoptysis could also suggest a secondary pulmonary infection that can be seen in patients with traction bronchiectasis and in those receiving immunosuppressive therapy. Chest pain is rare in most of the ILDs with the exception of sarcoidosis where chest discomfort is not uncommon. Fatigue is common to all of the ILDs.

Past Medical History The most pertinent history includes a personal history of a CTD or a history of symptoms commonly associated with a CTD (e.g., Raynaud's phenomena). It is also important to remember that ILD associated with a CTD can be the initial presenting symptom of the disease and can precede the development of additional symptomatology by many years. A history of malignancy is important because some malignancies can be associated with dermatomyositis-associated COP and sarcoid-like reactions. A history of asthma and allergic rhinitis might suggest a diagnosis of eosinophilic GPA.

AFKEBOOKS
SINCE 2013

Medications Many medications have been associated with ILD, and to complicate matters further, many medications commonly used to treat inflammatory and granulomatous lung disease are also associated with ILD development (e.g., methotrexate, azathioprine, rituximab, and the tumor necrosis factor α–blocking agents). Specific medications in many classes are also known to cause ILD, including antibiotics (e.g., nitrofurantoin), antiarrhythmics (e.g., amiodarone), and many of the antineoplastic agents (e.g., bleomycin).

Family History A family history of ILD (of almost any type) is important to ascertain. The percentage of pulmonary fibrosis that is familial, as opposed to idiopathic, varies by study, with estimates ranging from <5% to as high as 20%. Despite this variability, most agree that the presence of a close relative with an IIP is among the strongest risk factors for IPF. Family studies have consistently noted familial aggregation of diverse forms of IIP (such as IPF, nonspecific interstitial pneumonia [NSIP], and DIP running in the same family) and, in some cases, other forms of ILD. To date, the most well-replicated genetic factors for pulmonary fibrosis (a promoter variant of a mucin gene [*MUC5B*]) and various genetic determinants known to influence telomere length (e.g., variants in the telomerase reverse transcriptase gene [*TERT*]) (Chap. 482) appear to be associated with both familial and idiopathic forms of pulmonary fibrosis similarly.

Social History A history of smoking is nearly always present in some forms of ILD (e.g., respiratory bronchiolitis and desquamative interstitial pneumonia [DIP]—sometimes referred by pathologists jointly as smoking-related ILD) where it is felt to be causative. A history of smoking is also noted in approximately three-quarters of IPF patients. Occupational and environmental exposure histories are also important to obtain as they might identify exposures known to cause pulmonary fibrosis (e.g., significant asbestos exposure) or HP (pigeon breeder's lung).

PHYSICAL EXAMINATION

End-inspiratory fine crackles, or rales, noted at the lung bases are found in most patients with IPF and may be one of the earliest signs of the disease. However, rales are nonspecific and can be found in many forms of ILD and other disorders. Wheezing is uncommon in most forms of ILD but can be present in some disorders, such as sarcoidosis, HP, and eosinophilic GPA. Signs of advanced disease include cyanosis, digital clubbing, and cor pulmonale.

LABORATORY STUDIES

Laboratory studies can be particularly helpful in the workup for an underlying CTD-associated ILD. As noted previously, these tests can reveal the presence of an underlying CTD as the cause of an ILD (e.g., a positive anti-cyclic citrullinated peptide [anti-CCP] antibody for RA) even when no other symptomatology or physical examination findings suggestive of the disorder are present. However, the cost-effectiveness and the extent of laboratory testing that should be ordered in various clinical contexts have yet to be determined (as there is a relatively long list of autoantibody tests that could be ordered).

PULMONARY FUNCTION TESTS

Most forms of ILD will eventually result in a restrictive deficit on pulmonary function testing. A restrictive deficit is typified by a reduced total lung capacity (TLC) and symmetrically reduced measures of forced expiratory volume in 1 s (FEV_1) and forced vital capacity (FVC). A reduction in the diffusing capacity of the lung for carbon monoxide (D_{LCO}) is also common and may precede a reduction in lung volumes; however, there is more measurement variability in D_{LCO} measurement and the test is less specific for ILD. A reduced FEV_1 to FVC ratio, which is diagnostic of airway obstruction, is unusual in many forms of ILD but can be present as an isolated finding or in conjunction with an additional restrictive deficit in ILDs involving the airways such as sarcoidosis, HP, and LAM. Although pulmonary function testing is rarely diagnostic, reductions in lung function help to characterize the extent of disease, and evidence for decline in repeated measures of pulmonary function (e.g., FVC) has been correlated with an elevated rate of mortality.

CHEST IMAGING STUDIES

Chest X-Ray Findings on CXR can be the first clinical indication that an ILD might be present. For example, enlarged hilar lymph nodes and a pattern of central nodular opacities in the mid to upper lung zones can suggest sarcoidosis. A basilar reticular pattern, with small cystic spaces, in the absence of clinical evidence for heart failure, might suggest IPF. With a few exceptions, CXR alone rarely leads to a specific diagnosis.

Chest CT High-resolution CT (HRCT) chest imaging is now considered to be standard of care in the initial evaluation of a patient with a suspected ILD. HRCT can be diagnostic for some ILDs (e.g., IPF) in the right clinical context and may preclude the need for, and spare the patient the risk of, a lung biopsy. HRCT also helps to define the extent of the ILD, determine the presence of more concerning features suggestive of advanced disease (e.g., honeycombing), provide information on coexisting diseases (e.g., emphysema and lung cancer), and when not diagnostic, provide the most useful locations for obtaining lung biopsy specimens.

LUNG BIOPSY

Fiberoptic Bronchoscopy Bronchoscopy can be helpful in establishing a specific ILD diagnosis, and can help to establish an alternate diagnosis, in select cases. Examination of serial lavage fluid can be helpful in establishing DAH, which can be present in ILDs with vasculitis (e.g., GPA), and in some cases, cellular examination can suggest a specific diagnosis (eosinophilia >25% in chronic eosinophilic pneumonia or fat globules in macrophages in lipoid pneumonia). Transbronchial lung biopsies and lymph node biopsies (in sarcoidosis in particular) can lead to a confident diagnosis in patients with likely granulomatous lung disease (e.g., sarcoidosis and HP). However, in general, bronchoscopically obtained tissue samples are often felt to be insufficient to diagnose most of the IIPs. To date, studies have been mixed on whether bronchoscopically obtained cryobiopsies, which can result in yields larger than those obtained by transbronchial forceps biopsies, could improve the diagnostic yield of bronchoscopy; however, the precise role of cryobiopsies in the diagnostic workup of ILD has yet to be clarified.

Surgical Lung Biopsy A surgically obtained lung biopsy specimen can help solidify the diagnosis of ILD. In many cases, these are now obtained through a video-assisted thoracoscopic (VATS) approach (as compared to an open thoracotomy), which tends to reduce the length of operative times and hospital stays. The diagnostic yield of biopsies tends to be higher if obtained prior to treatment. The desire to obtain a surgical lung biopsy should be weighed against the risks, which can include a short-term mortality rate of as high as 5%. These risks are reported to be higher in biopsies of patients ultimately diagnosed with IPF and in those presenting acutely.

INDIVIDUAL FORMS OF ILD

The ILDs include a diverse group of lung pathologies that can be subclassified into those disorders of unknown cause (e.g., IIPs) and those of known cause (e.g., sometimes referred to as secondary interstitial pneumonias [CTD-associated ILDs]) (see Fig. 293-1). Although this remains a useful approach to classifying this diverse group of disorders, it is important to recognize that genetic studies are challenging this classic categorization. For example, numerous ILDs commonly listed as having an "unknown cause" have been determined to have significant genetic underpinnings (e.g., IPF and LAM), while the pathophysiologic processes that result in ILDs of "known cause" (e.g., CTD) remain incompletely understood. Diagnosis is based on combined information obtained from a patient's clinical presentation, measures of pulmonary function, imaging, immune serologies, and histopathology. It is important to remember that prognosis and treatment vary widely by disorder (and disease extent). In some cases, medical therapy that is felt to be effective for some ILDs has been proven to be harmful for others. Medical treatments range from immune modulators to antifibrotic

AFKEBOOKS
SINCE 2013

medications, whereas lung transplantation remains the standard of care for patients with advanced and rapidly progressive ILDs.

IDIOPATHIC INTERSTITIAL PNEUMONIAS

IDIOPATHIC PULMONARY FIBROSIS

Clinical Manifestations IPF is the most common ILD of unknown cause. Prevalence increases with age and is estimated at 50–200:100,000. IPF is commonly diagnosed in the fifth or sixth decade in life, affects men more than women, and is frequently associated with a history of smoking or other environmental exposures. IPF is a variably progressive disease that carries a poor prognosis with an estimated 50% 3- to 5-year survival.

HRCT Image Findings Chest CT findings include subpleural reticulation with a posterior basal predominance usually including more advanced fibrotic features, such as honeycombing and traction bronchiectasis. Collectively, these imaging findings are referred to as a UIP pattern. The presence of extensive ground-glass opacities, bronchovascular changes, micronodules, mosaic attenuation, or an upper lung predominance should raise suspicion for an alternative diagnosis (Fig. 293-2).

Histopathology Diagnostic VATS biopsy findings include subpleural reticulation associated with honeycomb changes and fibroblast foci (subepithelial collections of myofibroblasts and collagen). These fibrotic changes alternate with areas of preserved normal alveolar architecture consistent with temporal and spatial heterogeneity (Fig. 293-3). Collectively, these pathologic findings are referred to as UIP.

Treatment Historically, IPF was felt to be refractory to medical therapy with lung transplantation the only viable therapeutic option. This dogma changed in 2014 with large clinical trials that demonstrated that antifibrotic therapy (pirfenidone and nintedanib) can slow decline of lung function in IPF patients. Further meta-analyses have suggested that antifibrotic therapy may also improve survival. Trials now suggest that antifibrotic therapy may be broadly effective in other forms of progressive pulmonary fibrosis as well. In contrast, treatment with immunosuppression, which had been commonly prescribed to many IPF patients, has now been demonstrated (in some cases) to be associated with increased morbidity and mortality. Physical therapy and supplemental oxygen, when indicated, can improve exercise tolerance and reduce likelihood of developing pulmonary hypertension. Lung transplantation can extend survival and improve the quality of life in a subset of IPF patients who meet criteria to undergo transplant.

NONSPECIFIC INTERSTITIAL PNEUMONIA

Clinical Manifestations Idiopathic NSIP is a distinct clinical entity with characteristic clinical, radiologic, and pathologic features; however, NSIP is also commonly observed in patients with CTD and less frequently with familial interstitial pneumonia, drug toxicity, and infection. Although the prevalence of NSIP is not well established, it is

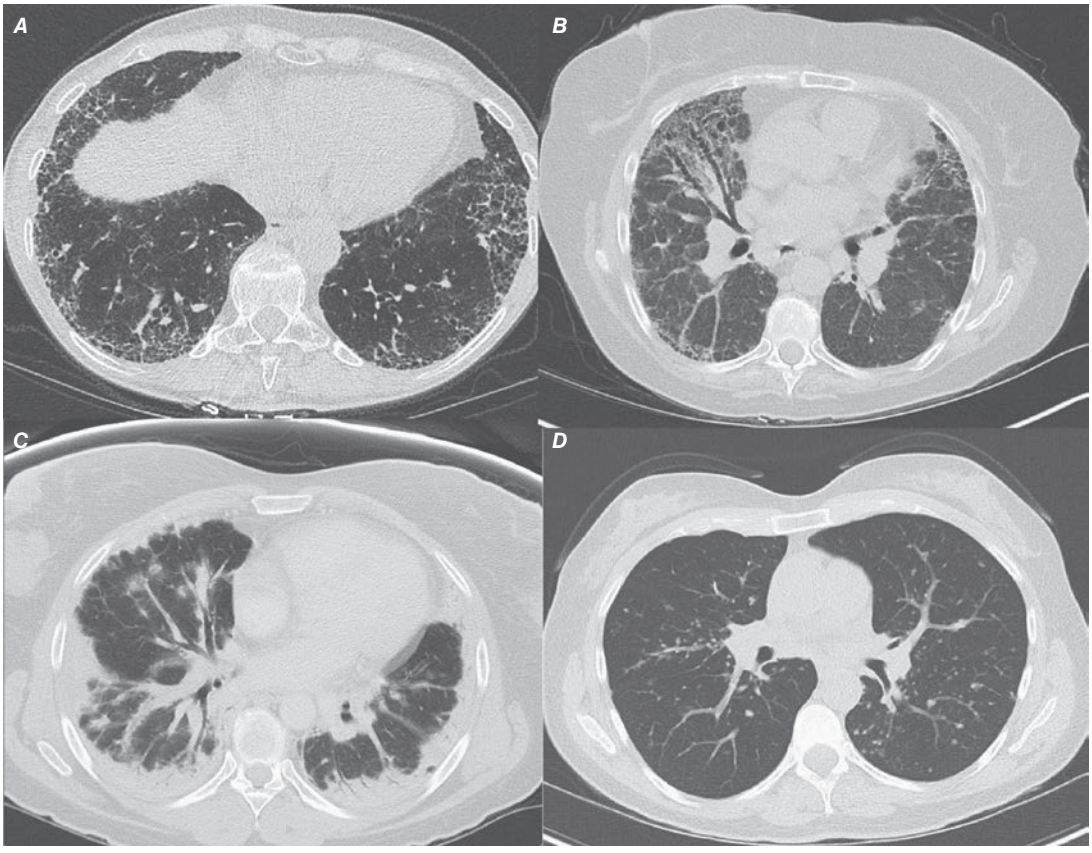


FIGURE 293-2 Chest CT imaging and interstitial lung disease. ***A.*** Idiopathic pulmonary fibrosis (IPF): Classic findings of IPF (apparent on this image) include a posterior, basilar predominance of subpleural reticular markings and more advanced features of pulmonary fibrosis including traction bronchiectasis and honeycombing. This constellation of findings is often referred to as a usual interstitial pneumonia (UIP) pattern. ***B.*** Nonspecific interstitial pneumonia (NSIP): Chest CT findings of NSIP can overlap with those of a UIP pattern but tend to include a bilateral, symmetric pattern that presents with a greater percentage of ground-glass opacities than is apparent in a UIP pattern. Additional unique findings include more diffuse imaging abnormalities with a predominance not limited to the lung bases, imaging abnormalities that spare the subpleural regions, and thickening of the bronchovascular bundles (as is apparent in the right mid lung zone on this image). ***C.*** Cryptogenic organizing pneumonia: Chest CT findings include patchy, sometimes migratory, subpleural consolidative opacities (as is apparent on this image) often with associated ground-glass opacities. Peribronchiolar or perilobar opacities can be present, and sometimes a rim of subpleural sparing (often referred to as a reversed halo or atoll sign) can be seen, which can help to aid in the diagnosis. ***D.*** Sarcoidosis: Sarcoidosis can present with varied imaging abnormalities, but a pattern of mediastinal and hilar lymphadenopathy with a pattern of reticular-nodular opacities involving the bronchovascular bundles (apparent in this image) are common features. Additional findings can include diffuse small nodules in a miliary pattern, larger nodular opacities, extensive ground-glass infiltrates, and mosaic attenuation suggestive of small airways involvement, and, in more advanced cases, signs of pulmonary fibrosis.

AFKEBOOKS
SINCE 2013

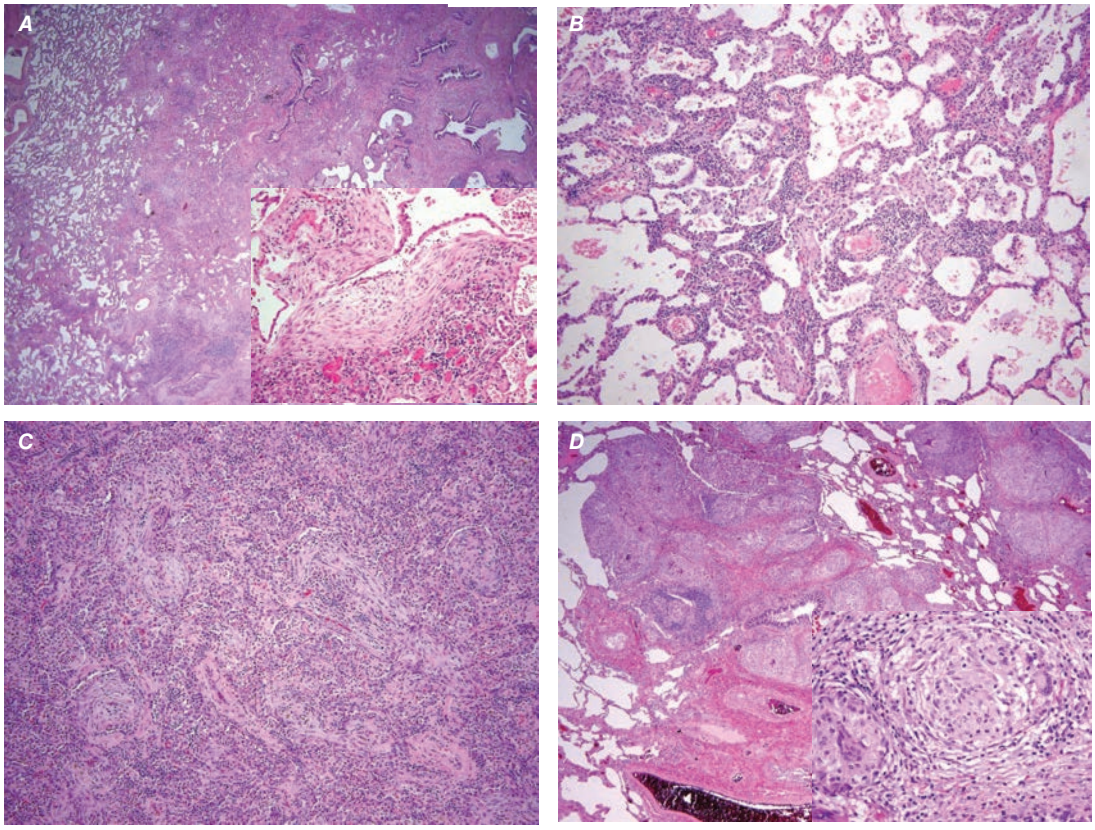


FIGURE 293-3 Histopathology of interstitial lung disease. ***A.*** Idiopathic pulmonary fibrosis (IPF): Histopathologic findings include subpleural reticulation associated with honeycomb changes alternating with areas of preserved normal lung architecture referred to as temporal and spatial heterogeneity (as is apparent in the low-power image above). Additional important diagnostic findings include fibroblast foci, which are subepithelial collections of myofibroblasts and collagen (as is apparent in the higher-powered inset of this image). Collectively, these pathologic findings are referred to as usual interstitial pneumonia (UIP). ***B.*** Nonspecific interstitial pneumonia (NSIP): Histopathologic findings of NSIP include varying amounts of interstitial inflammation and fibrosis with a uniform appearance (as is apparent in this image). Honeycomb changes are usually absent and fibroblast foci are rare. NSIP is often referred to histopathologically as being either predominantly cellular or fibrotic. ***C.*** Cryptogenic organizing pneumonia (COP): Histopathologic findings of COP include patchy regions of organizing pneumonia with granulation tissue that commonly involves the small airways, alveolar ducts, and alveoli with surrounding inflammation that can involve the alveolar walls (as is apparent in this image). ***D.*** Sarcoidosis: The hallmark histopathologic feature of sarcoidosis is presence of granulomas (as are apparent numerously in the low-powered image and more closely visualized in the higher-powered inset image). Typically, these are referred to as noncaseating, which suggests the absence of necrosis. Caseating granulomas are rare in sarcoid and should prompt additional evaluation for an underlying infection. Because malignancy can result in a granulomatous reaction, it is important to closely survey biopsy specimens with granulomatous involvement for additional signs of malignancy.

commonly diagnosed in nonsmoking females in their fifth decade of life. Positive serologic tests for CTD are frequently observed. Idiopathic NSIP has a relatively good prognosis, with a 5-year survival of >80%; patients with a predominant cellular NSIP pattern have a more favorable prognosis than those with a fibrosing NSIP pattern.

HRCT Image Findings Diffuse subpleural, symmetric, ground-glass, and reticular opacities are common. Volume loss and traction bronchiectasis involving the lower lung zones can also be found. Occasionally subpleural sparing is noted, while peribronchiolar thickening and honeycombing are uncommon.

Histopathology Diagnostic lung biopsy findings include varying amounts of interstitial inflammation and fibrosis with a uniform appearance. Honeycomb changes are usually absent and fibroblast foci are rare. NSIP is often referred to histopathologically as being either predominantly cellular (and potentially more responsive to medical therapy) or fibrotic (and potentially less likely to resolve with medical therapy).

Treatment Pulmonary fibrosis associated with CTD is commonly treated with immunosuppression despite the paucity of randomized clinical trials to demonstrate efficacy. Idiopathic NSIP is often treated with oral steroids (prednisone), cytotoxic agents (mycophenolate, azathioprine, and cyclophosphamide), or biologics (rituximab). Trials now suggest that NSIP patients with progressive pulmonary fibrosis may benefit from antifibrotic therapy. Oxygen therapy, pulmonary rehabilitation, and lung transplantation may be required in patients with progressive disease.

SMOKING-RELATED ILD

Although smoking-related ILDs, including respiratory bronchiolitis with interstitial lung disease (RB-ILD), and DIP are frequently subclassified with the IIPs, these disorders (along with PLCH, an ILD with unique clinical, imaging, and histopathologic manifestations) are commonly felt to be the result of active or prior tobacco smoke exposure. DIP has also been known to occur in children with familial pulmonary fibrosis (FPF). Smokers, particularly elderly smokers, frequently have radiologic (centrilobular) interstitial abnormalities. These interstitial abnormalities are often incidentally found on routine CXR or chest CT studies in asymptomatic or minimally symptomatic individuals. Respiratory bronchiolitis is felt to correlate histopathologically with these imaging findings. However, in some cases, these imaging findings can progress to more advanced radiologic changes where more diffuse signs of interstitial pneumonia tend to be present.

Clinical Manifestations These disorders predominantly occur in active, and in many cases heavy, smokers who are typically between 40 and 50 years of age. In those ultimately diagnosed with RB-ILD or DIP, dyspnea and cough are relatively common and symptomatic wheezing is not rare. The prevalence of smoking-related ILDs is not well understood, but they are generally felt to account for <10% of the IIPs. While there are minimal data on the natural histories and prognoses of these conditions, prolonged survival can be expected in most patients with RB-ILD and death secondary to progressive ILD is felt to be rare.

HRCT Image Findings Prominent and common findings in RB-ILD include central bronchial wall thickening, peripheral bronchial

AFKEBOOKS
SINCE 2013

wall thickening, centrilobular nodules, and ground-glass opacities. Septal lines and a reticular pattern are also not uncommon. Honeycombing is generally felt to be rare (and indicates a worse prognosis). Similar findings are noted in patients with DIP where diffuse (or patchy) bilateral symmetric ground-glass opacities tend to be even more prominent.

Histopathology Common features of RB-ILD include the accumulation of pigmented macrophages within the lumens of respiratory bronchioles and alveolar ducts, accompanied by chronic inflammation of the respiratory bronchiolar walls and both bronchiolar and peribronchiolar alveolar fibrosis causing architectural distortion. These features are patchy and confined to the peribronchiolar region. DIP tends to include similar changes but has a more diffuse pattern characterized by pigmented macrophage accumulation, pneumocyte hyperplasia, and prominent interstitial thickening.

Treatment All patients with smoking-related ILD should be counseled to discontinue smoking and/or encouraged to enroll in a formal smoking cessation program. Small studies have evaluated, and patients are often treated with, immunosuppressive (e.g., prednisone) and cytotoxic (e.g., azathioprine, and cyclophosphamide) agents and, in some cases, bronchodilators. To date, there is no strong evidence that these therapies result in significant improvements in symptoms or measures of pulmonary function or prevent clinical deterioration.

CRYPTOGENIC ORGANIZING PNEUMONIA

Clinical Manifestations COP typically involves patients in their 50–60s and often presents as a subacute flulike illness, with cough, dyspnea, fever, and fatigue. Inspiratory rales are often present on examination, and most patients are noted to have restrictive lung deficits on pulmonary function testing with hypoxemia. COP is commonly mistaken for pneumonia. It is important to note that this syndrome can occur in isolation, can be secondary to an underlying CTD (e.g., polymyositis) or medications, or can result from an underlying malignancy. Laboratory testing for various CTDs is helpful as testing can both be diagnostic and suggest the need for prolonged medical therapy.

HRCT Image Findings The most common imaging findings include patchy, sometimes migratory, subpleural consolidative opacities often with associated ground-glass opacities. Peribronchiolar or perilobar opacities can be present, and sometimes a rim of subpleural sparing (often referred to as a reversed halo or atoll sign) can be seen, which can aid in the diagnosis.

Histopathology Surgical lung biopsy specimens tend to reveal patchy regions of organizing pneumonia with granulation tissue that commonly involves the small airways, alveolar ducts, and alveoli with surrounding inflammation that can involve the alveolar walls (see Fig. 293-3).

Treatment Corticosteroids can result in substantial clinical improvement in many patients but usually need to be continued for at least 6 months as relapse rates are high. Evidence is growing that alternate cytotoxic (e.g., mycophenolate, cyclophosphamide) or biologic (e.g., rituximab) therapies can be helpful in both treating the disease and reducing the need for steroids. In some patients with secondary forms of the disease, long-term therapy may be needed.

ACUTE OR SUBACUTE IIPS

ACUTE INTERSTITIAL PNEUMONIA (HAMMAN-RICH SYNDROME)

Clinical Manifestations AIP is a rare and often fatal lung disorder that is characterized by an acute onset of respiratory distress and hypoxemia. A prodromal period of symptoms consistent with an acute upper respiratory infection is common. The mortality rate within 6 months of presentation can be quite high (>50%), and recurrences are common. In those who recover, lung function improvement can be substantial. AIP can be difficult to distinguish from acute respiratory distress syndrome (ARDS) and an acute exacerbation of an unsuspected underlying pulmonary fibrotic process.

HRCT Image Findings The most common imaging findings are patchy bilateral ground-glass opacities. Dependent regions of air-space consolidation are also common.

Histopathology Similar to ARDS and acute exacerbations of underlying pulmonary fibrosis, AIP presents histopathologically as diffuse alveolar damage (DAD) demonstrated on a surgical lung biopsy.

Treatment Treatment is mostly supportive and often includes mechanical ventilation. There is no proven drug therapy for AIP. Glucocorticoids are often given, but they are not clearly effective and data on their use in other forms of DAD (e.g., ARDS) is controversial.

ACUTE EXACERBATIONS OF IIPS

Clinical Manifestations Acute exacerbations are not separate disorders, but rather an accelerated phase of lung injury that can occur in any ILD resulting in pulmonary fibrosis. Acute exacerbations are most commonly described and most severe in patients with known IPF. Acute exacerbations are characterized by an acute onset (<30 days) of respiratory distress and hypoxemia occurring in a patient with underlying pulmonary fibrosis not explained by an alternate cause (e.g., pneumonia, left heart failure). Reported mortality rates are very high (>85%), and mean survival periods range from as little as days to months.

HRCT Image Findings The most common imaging findings include patchy bilateral ground-glass opacities and dependent regions of air-space consolidation. Sometimes these new changes can be appreciated on the background of the imaging findings typified by the underlying IIP, although sometimes they obscure the preceding imaging findings.

Histopathology Acute exacerbations of underlying pulmonary fibrosis present histopathologically as DAD, although sometimes organizing pneumonia can also be demonstrated on a surgical lung biopsy.

Treatment Treatment is mostly supportive. Mechanical ventilation, when not being used as a bridge to lung transplantation, is controversial as the survival rate in these patients tends to be poor. There is some evidence that drug therapy (e.g., nintedanib) may reduce the rate of acute exacerbations in patients with IPF. Drug therapy, in the context of an acute exacerbation, is also controversial. Immunosuppressive (e.g., prednisone) and cytotoxic (e.g., cyclophosphamide) therapies are commonly used without proven benefit.

ILD ASSOCIATED WITH CONNECTIVE TISSUE DISEASE

ILD is a common disease manifestation of many CTDs. Disease progression, response to therapy, and survival are variable and associated with specific radiologic and histopathologic patterns. ILD occurs most commonly in patients with scleroderma (systemic sclerosis form, or SSc), RA, polymyositis/dermatomyositis, and less frequently Sjögren's syndrome and systemic lupus erythematosus (SLE). ILD may precede the development of extrapulmonary manifestations of a specific CTD or may present as part of a poorly defined CTD. In rare cases, lung manifestations may be the sole feature of the patient's clinical presentation.

SYSTEMIC SCLEROSIS

Clinical Manifestations (Chap. 360) ILD is the most common pulmonary manifestation of SSc. ILD occurs in ~50% of SSc patients with diffuse disease and in ~30% of patients with limited disease. Pulmonary hypertension can occur separately or concomitantly with ILD and is more frequent in patients with limited SSc.

HRCT Image Findings Similar imaging findings noted in both patients with NSIP and IPF can be present, although findings consistent with COP and DAD may also be present. Additional HRCT findings may include a dilated esophagus and pulmonary artery enlargement.

AFKEBOOKS
SINCE 2013

Histopathology Comparable to the imaging overlap, histopathologic changes commonly noted in patients with NSIP and IPF are frequently identified. Additionally, aspiration related to esophageal dysmotility is common in SSc, and in these patients, histopathologic findings consistent with COP and DAD may be observed.

Treatment Cyclophosphamide has a modest benefit in preservation of lung function and is associated with significant toxicity. Mycophenolate has recently been shown to have similar efficacy and improved tolerability. Clinical trials have demonstrated that antifibrotic therapy (e.g., nintedanib) may benefit patients with systemic sclerosis associated pulmonary fibrosis. Minimizing the risk of reflux by using high-dose proton pump inhibitors or antireflux surgery should be considered in SSc with progressive ILD. Lung transplantation can potentially be offered to select patients without significant aspiration or chest wall restriction.

RHEUMATOID ARTHRITIS

Clinical Manifestations (Chap. 358) A common extraarticular complication of RA is ILD. Although RA is more common in females, RA-ILD is more frequent in males and in patients with a history of tobacco exposure. In a small subset of patients, ILD is the first disease manifestation of RA. Clinically evident RA-ILD occurs in nearly 10% of the RA population; however, up to 40–50% of RA patients have radiologic abnormalities on chest CT, suggesting that ILD in the context of RA may be underdiagnosed.

HRCT Image Findings The most common imaging pattern of ILD in patients with RA is a UIP pattern, although NSIP patterns are not uncommon. There is evidence that survival in patients with RA is decreased in patients with a UIP pattern and among those with more extensive fibrosis in general.

Histopathology Histopathologic findings of UIP and NSIP are most common. Some studies suggest that UIP in the context of RA (as compared to IPF) may present with a reduced number of fibroblastic foci and an increased amount of germinal centers. Comparable to the imaging findings, UIP (and DAD) patterns in patients with RA are associated with reduced survival.

Treatment In contrast with SSc, there are no randomized clinical trials testing the role of immune suppression in RA-ILD. Extrapolating from the scleroderma experience, immunosuppressive (e.g., prednisone) and cytotoxic (e.g., mycophenolate, azathioprine, cyclophosphamide, and calcineurin inhibitors) agents have been used with variable success. Clinical trials testing antifibrotic therapies (pirfenidone and nintedanib) are presently being conducted. Lung transplantation is a viable therapeutic approach for eligible patients with progressive disease that is not responsive to medical therapy.

DERMATOMYOSITIS/POLYMYOSITIS

Clinical Manifestations (Chap. 365) The idiopathic inflammatory myopathies are disorders characterized by immune-mediated destruction and dysfunction of muscle; however, these disorders can affect the skin, joints, cardiovascular system, and lung. The prevalence of ILD associated with inflammatory myopathy varies by report; however, ILD is present in up to 45% of patients with positive anti-synthetase antibodies. The anti-synthetase syndrome is characterized by positive anti-synthetase antibodies, myositis, fever, Raynaud's phenomenon, mechanic's hands, arthritis, and progressive ILD. There is a subset of anti–Jo-1 antibody–positive individuals who can develop a rapidly progressive form of ILD consistent with an acute exacerbation. Some studies have suggested that ILD may be even more common in those with other antibodies (e.g., anti-PL-12). Dermatomyositis/polymyositis can occur as an isolated CTD or as a process associated with an underlying malignancy.

HRCT Image Findings Common imaging patterns of ILD in patients with dermatomyositis/polymyositis include those consistent with NSIP with or without evidence for COP. A UIP pattern can also occur. Some studies have suggested that a UIP pattern may be more common among those with anti-PL-12 antibodies.

Histopathology The anti-synthetase syndrome is associated with multiple histopathologic subtypes including NSIP, COP, and UIP. DAD, a histopathologic pattern observed in AIP and acute exacerbations, is associated with rapidly progressive ILD in myositis patients.

Treatment Immunosuppressive (e.g., prednisone) and cytotoxic (e.g., mycophenolate, azathioprine, cyclophosphamide, and calcineurin inhibitors) agents are often used in patients with progressive ILD. Some patients (particularly those with less fibrosis) have been noted to have improved or resolved ILD in response to medical therapy. In small studies, relapses have been more common in patients treated with prednisone alone. Patients who fail immune-suppressive therapy can benefit from lung transplantation.

GRANULOMATOUS ILDS

The most common granulomatous ILD is sarcoidosis, a multisystem disorder of unknown cause where lung involvement is often the most dominant feature; sarcoidosis is discussed in **Chap. 367**. HP, a granulomatous reaction due to inhalation of organic (e.g., bird fancier's lung secondary to exposure to bird feathers) and inorganic (e.g., coal worker's pneumoconiosis secondary to exposure to coal dusts) dusts, is also an important and common cause of ILD and is discussed in **Chap. 288**.

Granulomatous Vasculitides (See Chap. 64) These disorders are characterized by blood vessels with inflammatory infiltrates and associated granulomatous lesions with or without the presence of tissue necrosis. The lungs are commonly involved, and a unique feature of these disorders is that hemoptysis can be a presenting symptom. Although laboratory testing is often helpful and can provide specific information, biopsies of involved tissue can be essential for making the diagnosis. Many of these disorders include additional systemic manifestations. GPA, also referred to as Wegener's disease, is an example of a granulomatous vasculitis that commonly affects the lung (including inflammatory infiltrates in small to medium-sized vessels), ears, nose, throat, and kidney (resulting in glomerulonephritis). Common imaging abnormalities of GPA include nodules, patchy ground-glass and consolidative opacities that can be migratory, and hilar lymphadenopathy. Eosinophilic GPA (EG; also referred to as Churg-Strauss syndrome) is another example of a granulomatous vasculitis that affects the lung (including eosinophilic infiltrates in small to medium-sized vessels) and can result in numerous clinical manifestations but frequently includes chronic sinusitis, asthma, and peripheral blood eosinophilia. Common imaging abnormalities of EG include peripheral consolidative opacities that can be migratory and small pleural effusions.

GENETICS AND ILD

Studies of genetic epidemiology have led to important insights in our understanding of ILD. First, studies of families with FPF have demonstrated that unique IIPs can cosegregate with specific genetic variants known to be associated with IPF. This suggests that many genetic variants appear to predispose to interstitial lung injury patterns more broadly than to unique diagnoses specifically. Second, most of the genetic variants known to be associated with FPF are also associated with more sporadic forms of the disease. Third, at least one of the genetic factors most strongly associated with FPF and IPF is both common and confers a large increase in the risk of these diseases. At least one copy of a mucin 5B (*MUC5B*) promoter variant is present in ~20% of Caucasian populations and 35–45% of patients with IPF and confers an approximate sixfold increase in the risk of this disease. Fourth, studies of general population samples demonstrate that imaging abnormalities suggestive of an early stage of pulmonary fibrosis in research participants without known ILD are not uncommon (occurring in ~7–9% of adults) and are also associated with the same genetic variants known to be associated with IPF (e.g., the *MUC5B* promoter variant). This latter finding suggests a path forward toward an early detection of IPF. Additional genetic findings demonstrating replicable associations with pulmonary fibrosis include numerous genetic variants in, and

AFKEBOOKS
SINCE 2013

adjacent to, genes known to be involved in the regulation of telomere length (e.g., the *TERT* gene, the telomerase RNA component [*TERC*] gene, and the regulator of telomere elongation helicase 1 [*RTEL1*] gene) and surfactant protein genes (e.g., surfactant protein A2 [*SFTPA2*] gene) **(Chap. 482)**.

Genetic studies have also provided some insights into other forms of ILD. Genome-wide association studies of sarcoidosis have demonstrated numerous variants in genes and in genomic regions that are associated with the disease. Some of these disease-associated variants in sarcoidosis fall in human leukocyte antigen (*HLA*) regions, in regions of genes involved in immune regulation (e.g., interleukin 12B [*IL12B*]), and in regions of genes that are less well understood (butyrophilin-like 2 [*BTNL2*]) but also appear to be involved in T-cell activation. LAM is often associated with genetic variants in the tuberous sclerosis complex genes (e.g., *TSC1* and *TSC2*), consistent with the known evidence that this disease can occur in isolation but also in patients with known tuberous sclerosis. Many genetic factors for rare diseases such as Hermansky-Pudlak syndrome (a rare autosomal recessive disorder that results in pulmonary fibrosis but also includes oculocutaneous albinism, bleeding diatheses, and horizontal nystagmus) have also been discovered (e.g., *HSP1*, and *HSP3-7*).

GLOBAL CONSIDERATIONS

The prevalence, clinical presentation, and natural history of most ILDs in European countries resemble those described in the United States. However, as expected, there is growing evidence for racial differences in clinical (rate of acute exacerbations) and genetic (*MUC5B*) attributes between Caucasian and Asian populations. To date, there are limited data on the prevalence of ILD in Hispanics, subjects of African descent, and many other ethnic groups.

FURTHER READING

American Thoracic Society/European Respiratory Society: Consensus classification of the idiopathic interstitial pneumonias. Am J Respir Crit Care Med 165:277, 2002.

Raghu G et al; ATS/ERS/JRS/ALAT Committee on Idiopathic Pulmonary Fibrosis: An official ATS/ERS/ JRS/ALAT statement: Idiopathic pulmonary fibrosis: Evidence-based guidelines for the diagnosis and management. Am J Respir Crit Care Med 183:788, 2011.

Travis WD et al: Idiopathic nonspecific interstitial pneumonia: Report of an American Thoracic Society project. Am J Respir Crit Care Med 177:1338, 2008.

Travis WD et al: An official American Thoracic Society/European Respiratory Society Statement: Ten decade update on IIP's, potential areas for future investigation are proposed (ATS/ERS update of the international multidisciplinary classification of the idiopathic interstitial pneumonias. Am J Respir Crit Care Med 188:733, 2013.

294 Disorders of the Pleura

Richard W. Light*

PLEURAL EFFUSION

The pleural space lies between the lung and the chest wall and normally contains a very thin layer of fluid, which serves as a coupling system. A pleural effusion is present when there is an excess quantity of fluid in the pleural space.

Etiology Pleural fluid accumulates when pleural fluid formation exceeds pleural fluid absorption. Normally, fluid enters the pleural space from the capillaries in the parietal pleura and is removed via the lymphatics in the parietal pleura. Fluid also can enter the pleural space from the interstitial spaces of the lung via the visceral pleura or from the peritoneal cavity via small holes in the diaphragm. The lymphatics have the capacity to absorb 20 times more fluid than is formed normally. Accordingly, a pleural effusion may develop when there is excess pleural fluid formation (from the interstitial spaces of the lung, the parietal pleura, or the peritoneal cavity) or when there is decreased fluid removal by the lymphatics.

Diagnostic Approach Patients suspected of having a pleural effusion should undergo chest imaging to diagnose its extent. Chest ultrasound has replaced the lateral decubitus x-ray in the evaluation of suspected pleural effusions and as a guide to thoracentesis. When a patient is found to have a pleural effusion, an effort should be made to determine the cause **(Fig. 294-1)**. The first step is to determine whether the effusion is a transudate or an exudate. A *transudative pleural effusion* occurs when *systemic factors* that influence the formation and absorption of pleural fluid are altered. The leading causes of transudative pleural

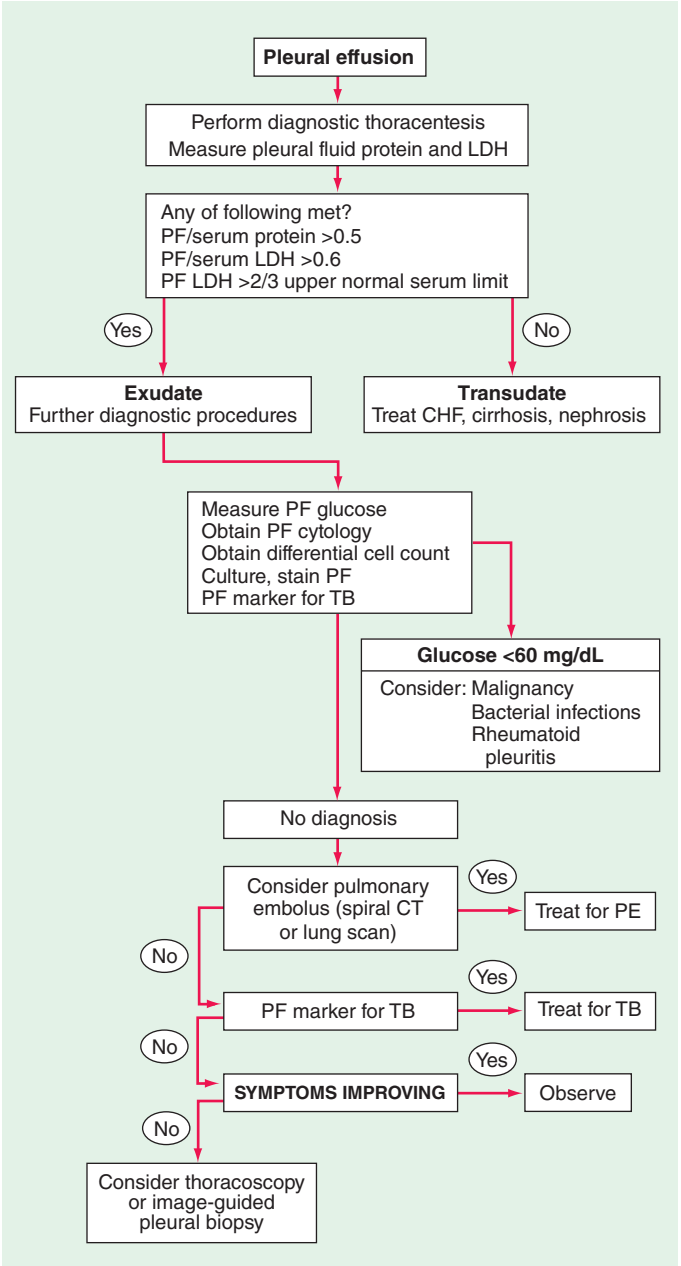


FIGURE 294-1 Approach to the diagnosis of pleural effusions. CHF, congestive heart failure; CT, computed tomography; LDH, lactate dehydrogenase; PE, pulmonary embolism; PF, pleural fluid; TB, tuberculosis.

*Deceased.